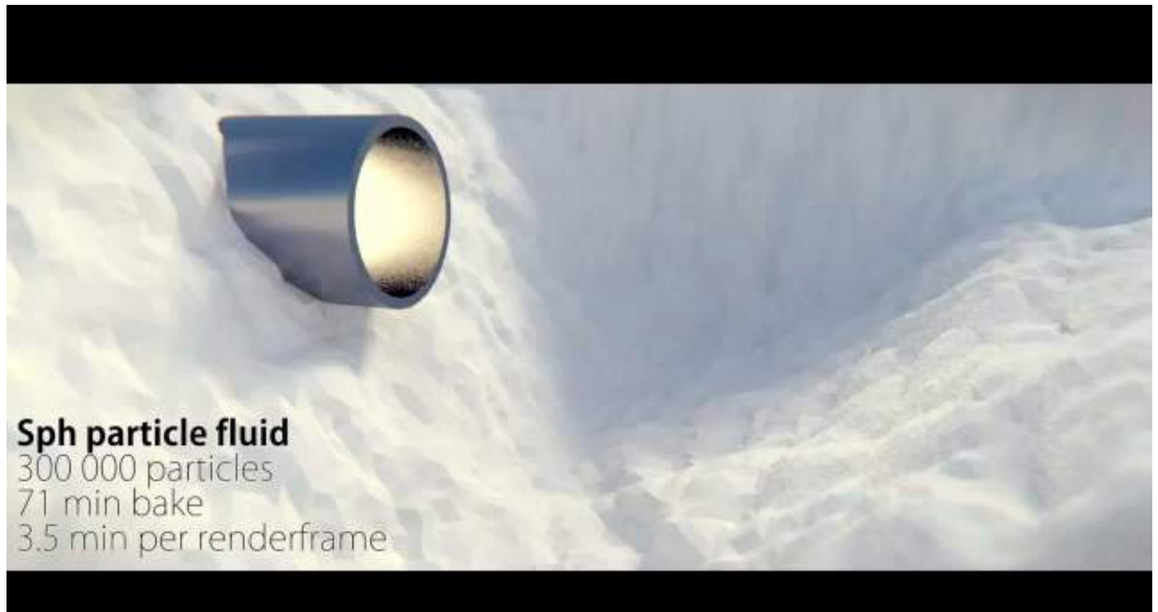
# Math (P)Review Part I:
# Linear Algebra

# Linear Algebra in Computer Graphics

- Today's topic: linear algebra.
- Why is linear algebra important for computer graphics?
  - Effective bridge between geometry, physics, etc., and *computation*.
  - In many areas of graphics, once you can express the solution to a problem in terms of linear algebra, you're essentially done: now ask the computer to solve Ax=b.
  - Fast numerical linear algebra has really made modern computer graphics possible (image processing, physically-based animation, geometry processing…)






# Vector Space—Formal Definition

- **Linear algebra is the study of vector spaces and linear maps between them—here's the formal definition*:**

For all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalars $a, b$:

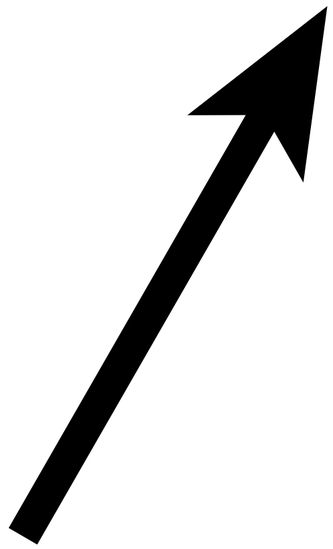
- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
- $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
- There exists a *zero vector* "$\mathbf{0}$" such that $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$
- For every $\mathbf{v}$ there is a vector "$-\mathbf{v}$" such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$
- $1\mathbf{v} = \mathbf{v}$
- $a(b\mathbf{v}) = (ab)\mathbf{v}$
- $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
- $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

- ***Where do these rules come from?***
- **In mathematics (and in life) you should never simply accept a set of rules handed to you by an authority…**
- **Let's try to understand where these "rules" come from.**

***this will NOT be on the test!**

# Vectors - Intuition

- First things first: what is a vector?
- Intuitively, a vector is a little arrow:

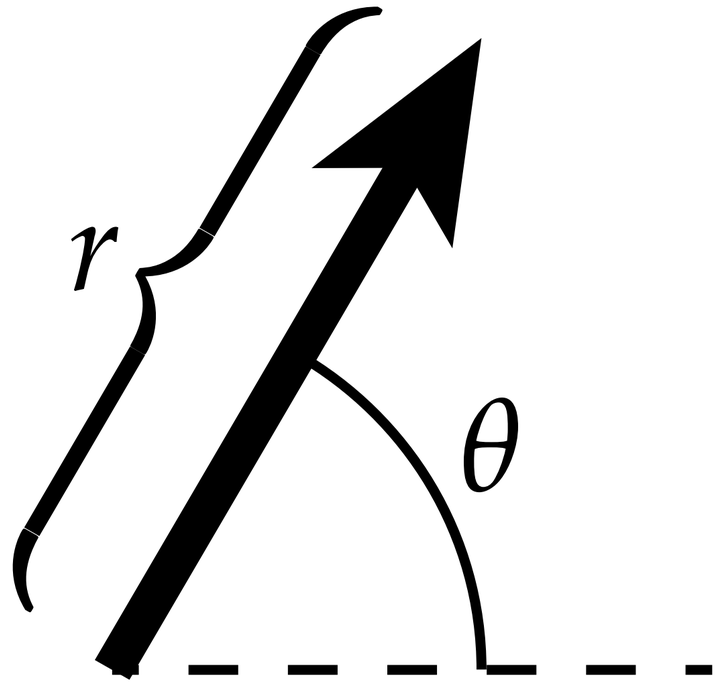


A vector.

- In computer graphics, we work with many types of data that may not look like little arrows (polynomials, images, radiance...). But they still *behave* like vectors. So, this little arrow is still often a useful mental model.

# Vectors - What Can We Measure?

- What information does a vector encode?
- Fundamentally, just direction and magnitude*:




- For instance, a vector in 2D can be encoded by a length and an angle relative to some fixed direction ("polar coordinates").
- (Side note: are these values the same in any coordinate system?)
- How else might we encode a vector?

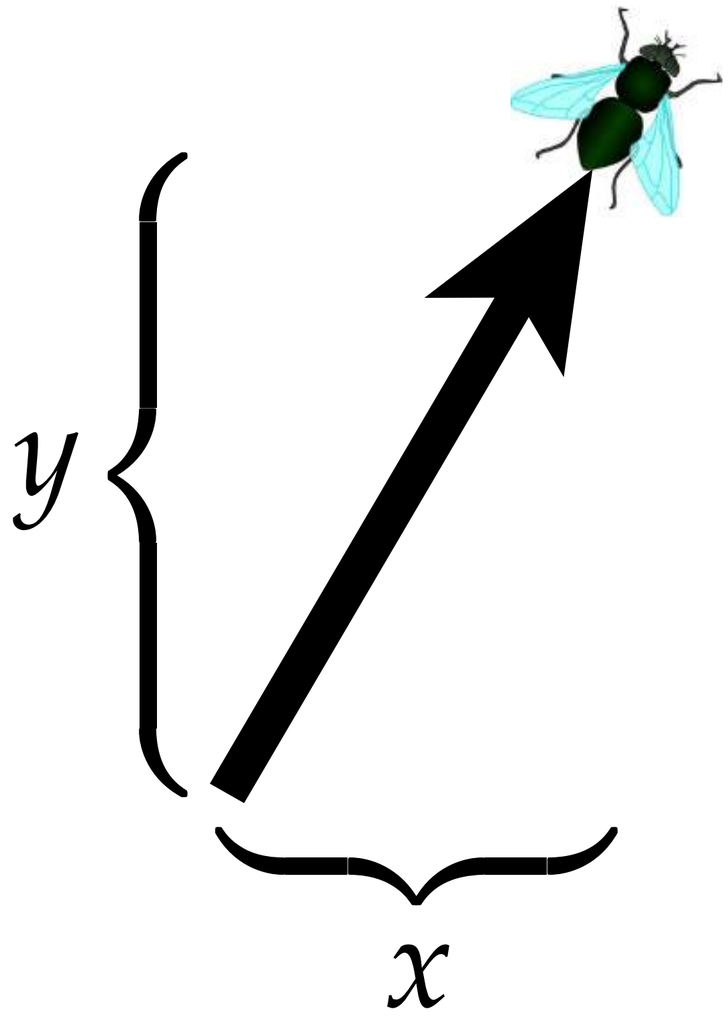
*Traditionally, a vector does not include a "basepoint"; a vector with a basepoint is sometimes called a *tangent vector*.

# Vector in Cartesian Coordinates

- Can also measure components of a vector with respect to some chosen *coordinate system*:



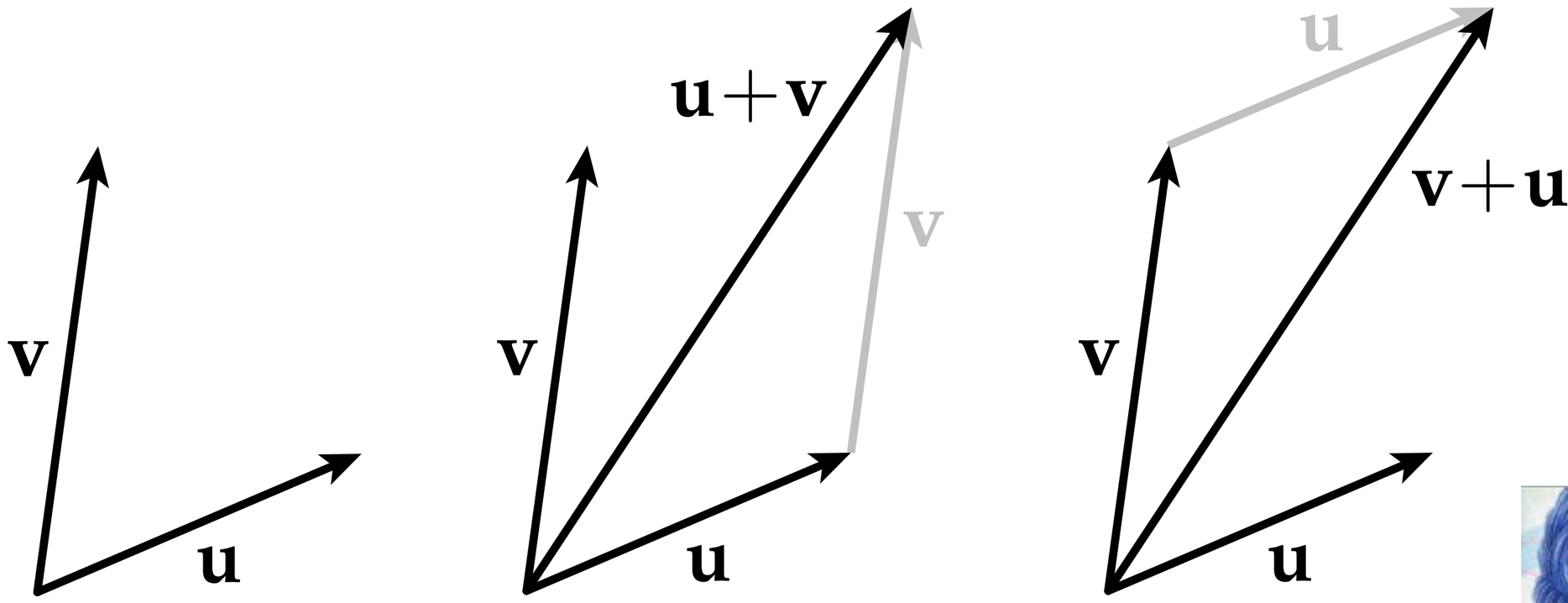
René Descartes, Est. 1596




- **WARNING**: Can't directly compare coordinates in different systems! (Also shouldn't compare $(r,\theta)$ to $(x,y)$.)

# What Can We Do with a Vector?

- Two basic operations. First, we can add them "end to end":





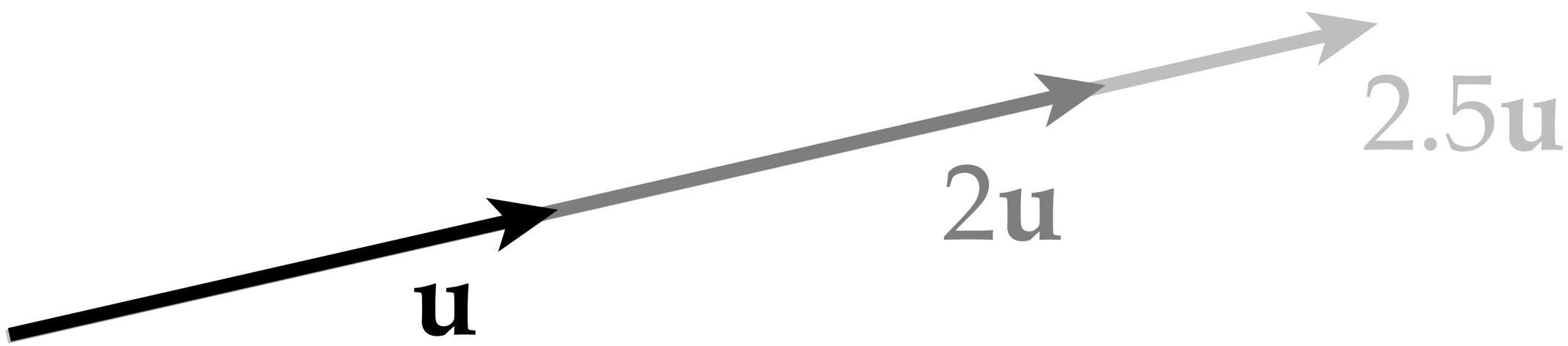


Niels Henrik Abel

- What if we walk along v first, then u?
- Actually, it doesn't seem to matter: u+v = v+u
- Language: vector addition is "commutative" or "abelian"

# What Else Can We Do with a Vector?

- Other basic operation? Scaling:

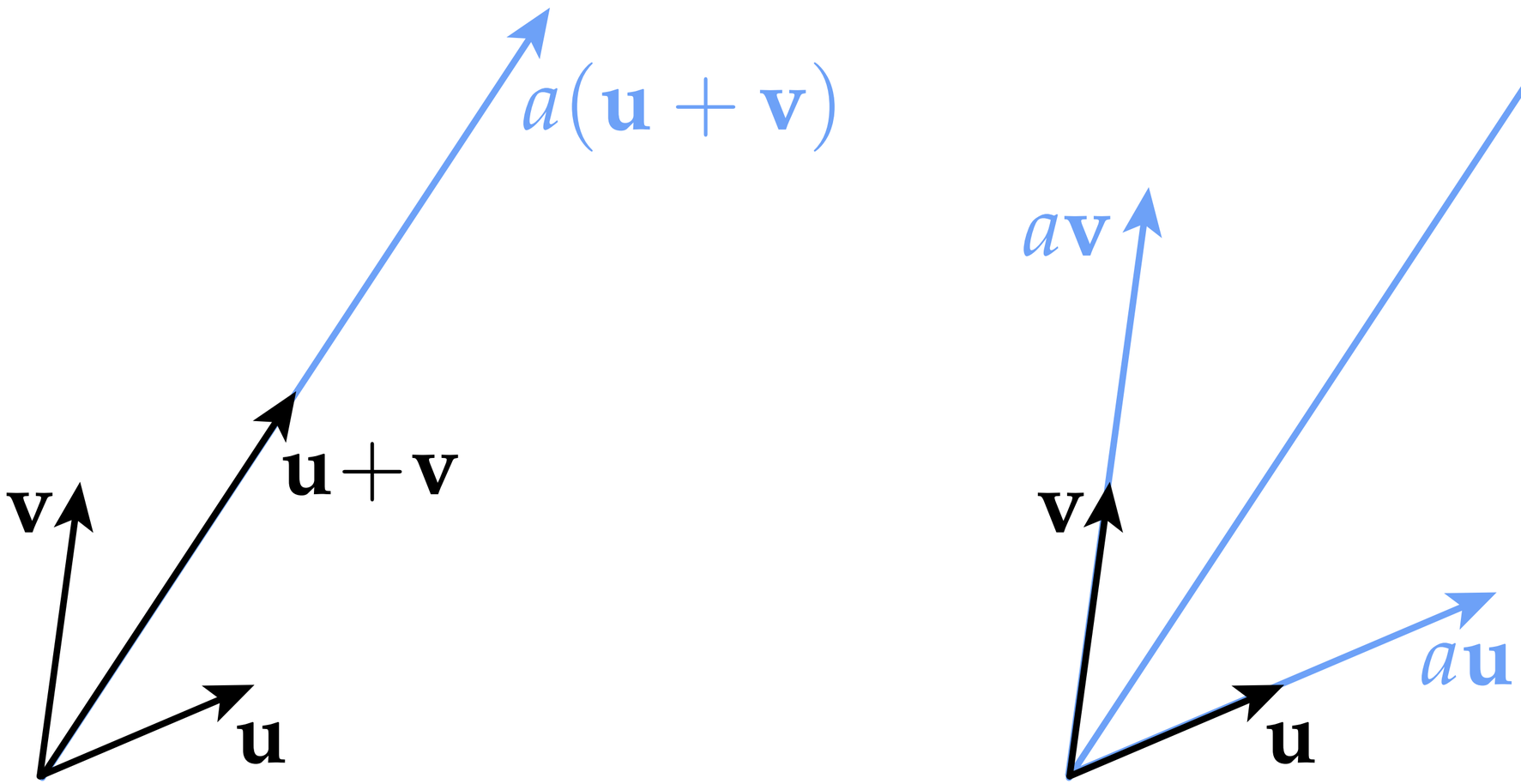



- In general, can multiply any vector **u** by a number or "scalar" $a$ to get a new vector $a\mathbf{u}$.
- Multiplication behaves the way we would expect, based on the geometric behavior of scaling "little arrows." E.g.,

$$a(b\mathbf{u}) = (ab)\mathbf{u}$$

# Interaction of Addition & Scaling

- What if we try to add two scaled vectors? Or scale two vectors that have been added together?




- Interesting—seems we get the same result either way:

$$a(\mathbf{u}+\mathbf{v}) = a\mathbf{u} + a\mathbf{v}$$

# Vector Space—Formal Definition

- **If we keep playing around vectors, eventually we come up with a complete set of "rules" that vectors seem to obey:**

  For all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalars $a, b$:

  - $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
  - $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
  - There exists a *zero vector* "$\mathbf{0}$" such that $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$
  - For every $\mathbf{v}$ there is a vector "$-\mathbf{v}$" such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$
  - $1\mathbf{v} = \mathbf{v}$
  - $a(b\mathbf{v}) = (ab)\mathbf{v}$
  - $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
  - $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

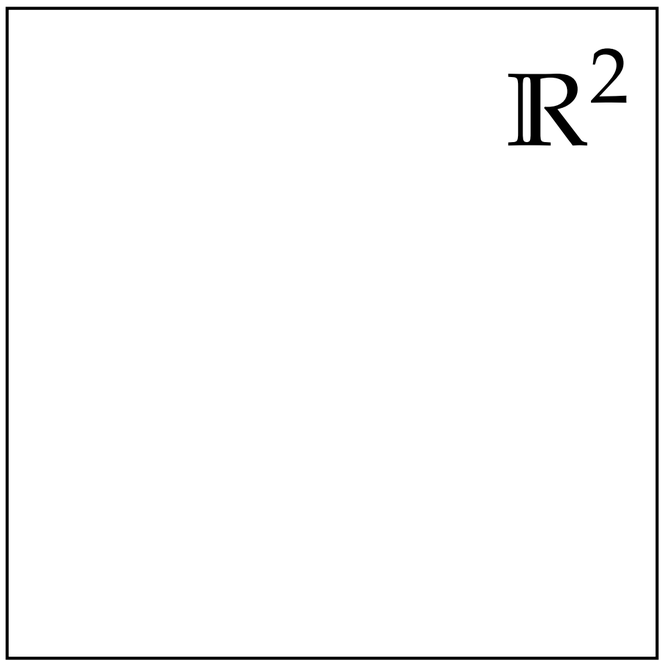
- ***These rules did not "fall out of the sky!"*** **Each one comes from the geometric behavior of "little arrows." (Can you draw a picture for each one?)**

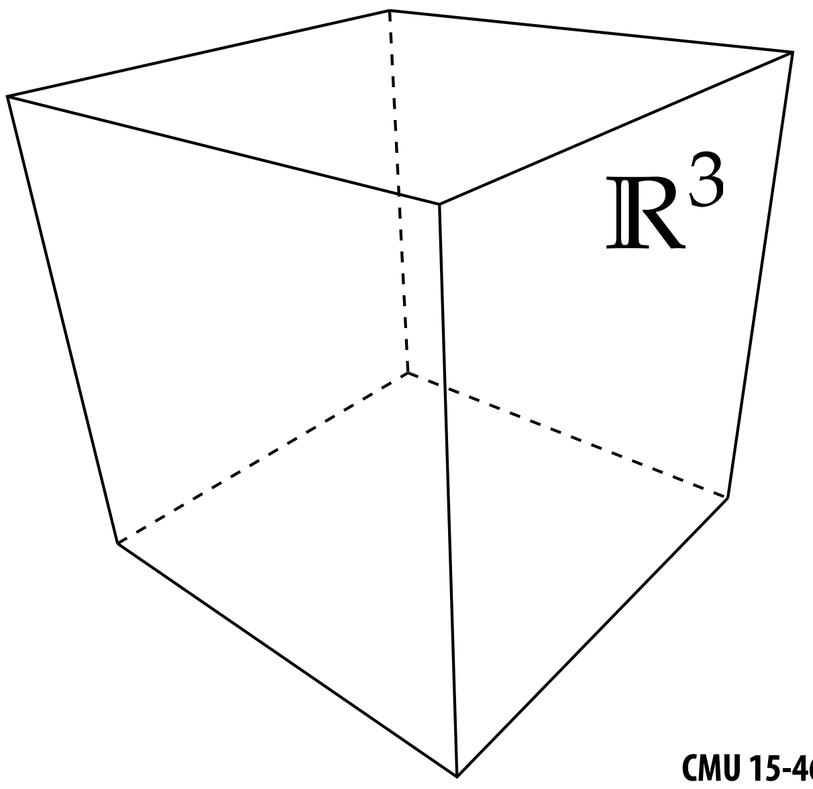
- ***Any*** **collection of objects satisfying all of this properties is a vector space (even if they don't look like little arrows!)**

# Euclidean Vector Space

- Most common example: Euclidean n-dimensional space
- Typically denoted by $\mathbb{R}^n$, meaning "n real numbers"
- E.g., (1.23, 4.56, π/2) is a point in $\mathbb{R}^3$
- Why such a common example?
  - Looks a lot like the space we live in!
  - That's what we can easily encode on a computer (a list of floating-point numbers).

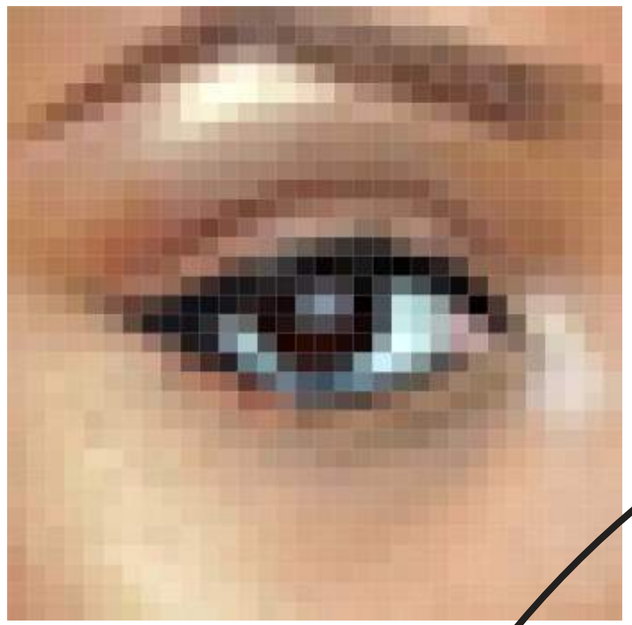


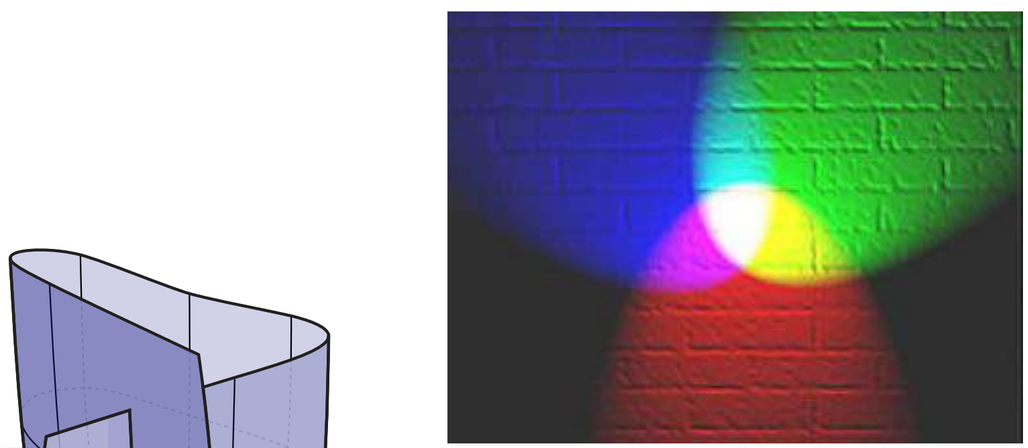



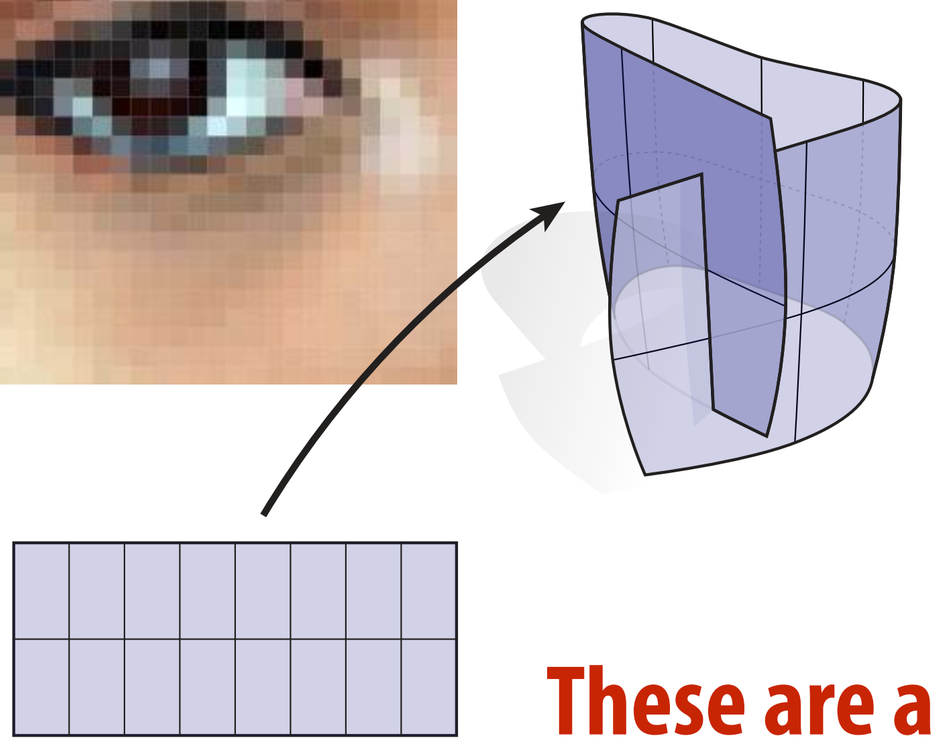



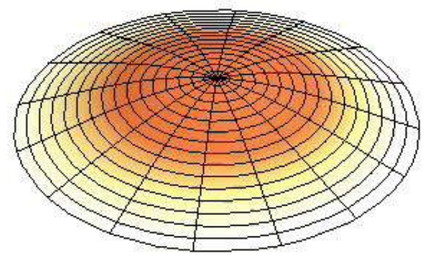
# Functions as Vectors

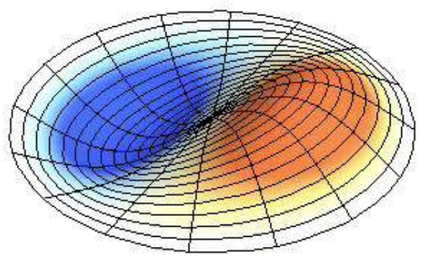
- Another very important example of vector spaces in computer graphics are spaces of *functions*.
- Why? Because many of the objects we want to work with in graphics are functions! (Images, radiance from a light source, surfaces, modal vibrations, ...)

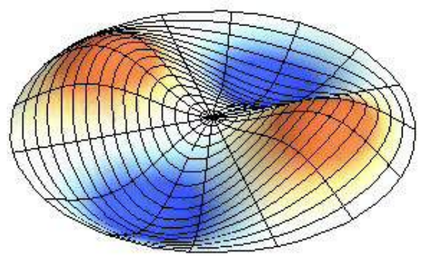










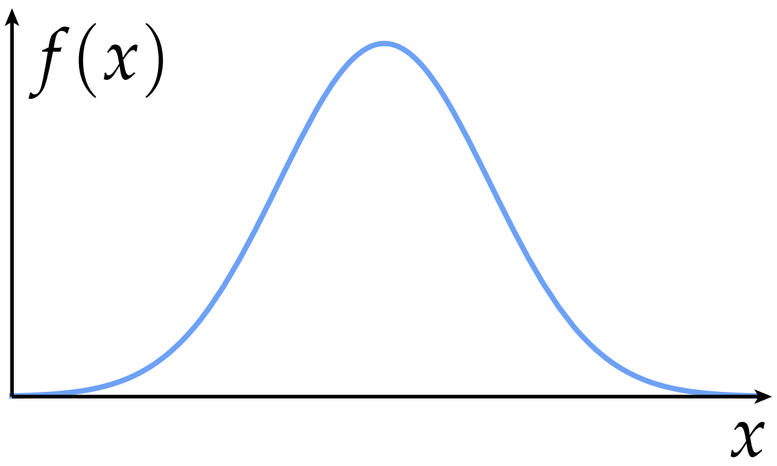


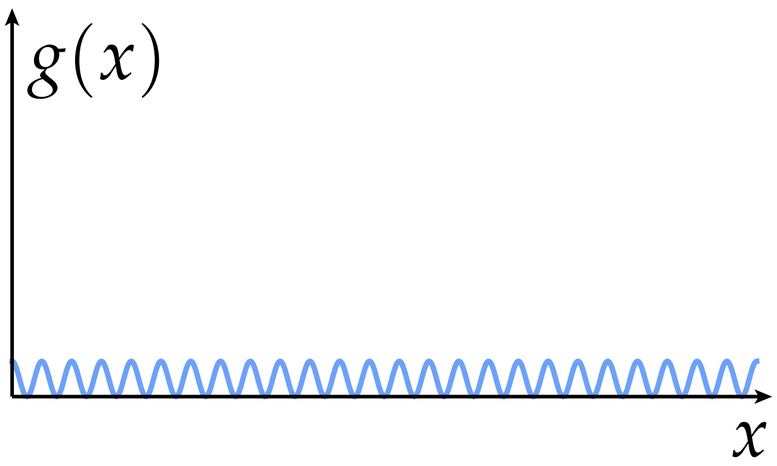
These are all vectors! :-)

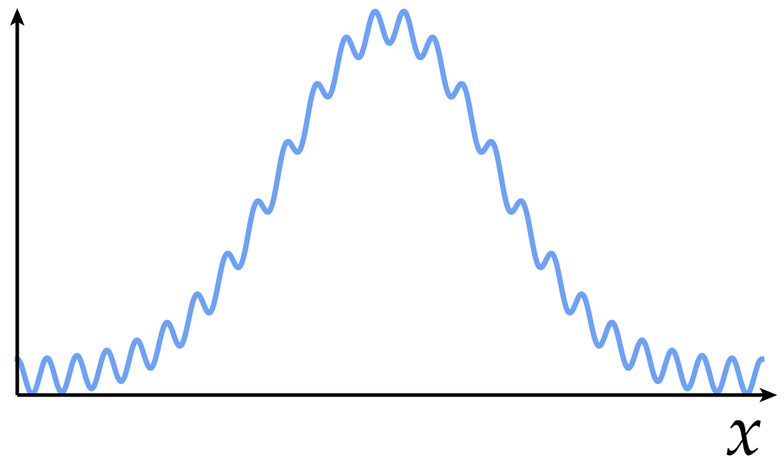
# Functions as Vectors

- Do functions exhibit the same behavior as "little arrows?"
- Well, we can certainly add two functions:

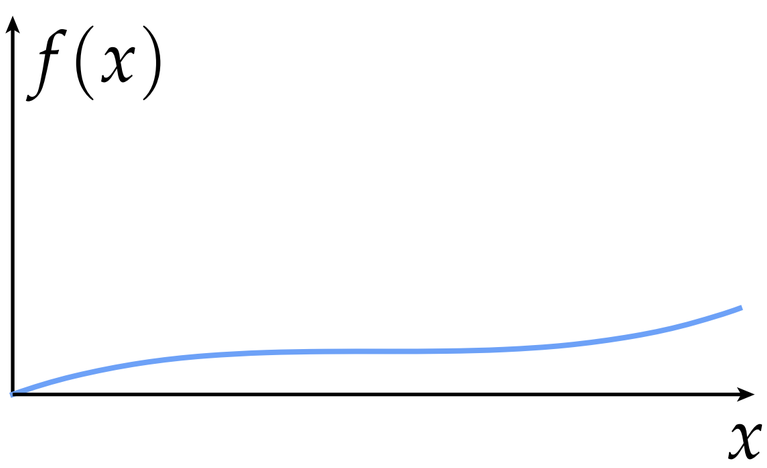
$$(f+g)(x) := f(x) + g(x)$$



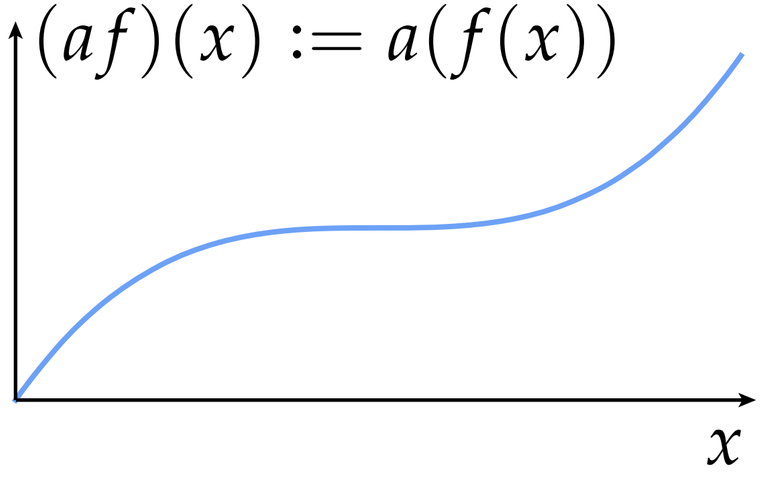




- We can also scale a function:




$$(af)(x) := a(f(x))$$



# Functions as Vectors

- **What about the rest of these properties?**

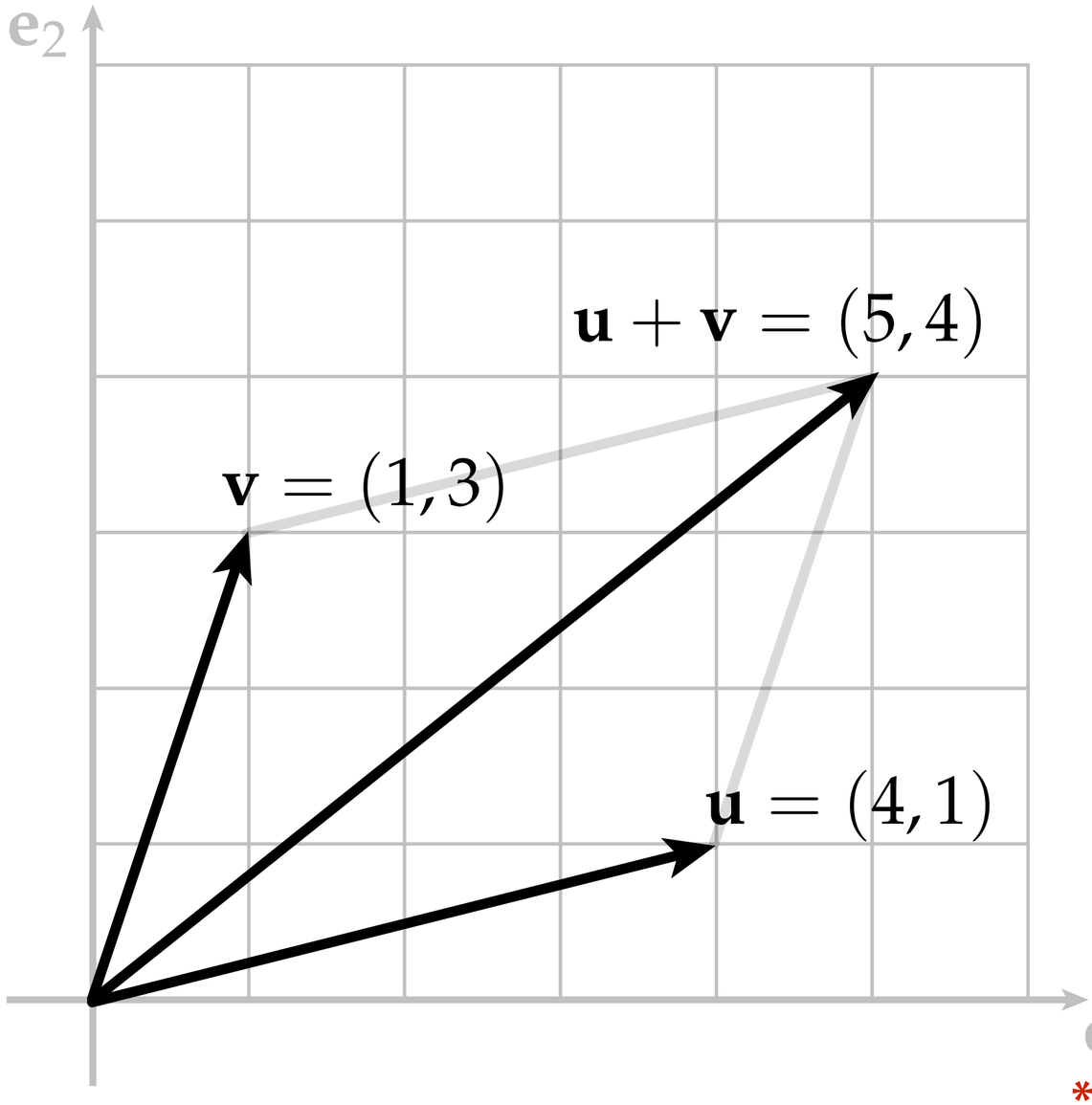
  For all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalars $a, b$:

  - $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
  - $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
  - There exists a *zero vector* "$\mathbf{0}$" such that $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$
  - For every $\mathbf{v}$ there is a vector "$-\mathbf{v}$" such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$
  - $1\mathbf{v} = \mathbf{v}$
  - $a(b\mathbf{v}) = (ab)\mathbf{v}$
  - $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + b\mathbf{v}$
  - $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

- **Try it out at home!**
- **E.g., the "zero vector" is the function equal to zero for all x.**
- **Short answer: yes, functions are vectors! (Even if they don't look like "little arrows".)**

# Vectors in Coordinates

- So far, we've only drawn our vector operations via pictures.
- How do we actually compute with vectors?
- Return to our coordinate representation:




$$\begin{aligned}\mathbf{u}+\mathbf{v} &= (1,3)+(4,1)\\ &= (1+4,3+1)\\ &= (5,4)\end{aligned}$$

*Side note: does it make sense to add vectors encoded as (r,θ)?

**Ok, so we came up with some rule for adding pairs of numbers.**

**How can we check that it faithfully encodes geometric behavior of "little arrows?"**

# From Geometry to Algebra

- **Just check that it agrees with our list of rules that we know (from reasoning *geometrically*) "little arrows" must obey:**

For all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalars $a, b$:

- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
- $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
- There exists a *zero vector* "$\mathbf{0}$" such that $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$
- For every $\mathbf{v}$ there is a vector "$-\mathbf{v}$" such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$
- $1\mathbf{v} = \mathbf{v}$
- $a(b\mathbf{v}) = (ab)\mathbf{v}$
- $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
- $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

- **For instance, for any two vectors $\mathbf{u} := (u_1, u_2)$ and $\mathbf{v} := (v_1, v_2)$ we have**

$$\mathbf{u} + \mathbf{v} = (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2) =$$
$$(v_1 + u_1, v_2 + u_2) = (v_1, v_2) + (u_1, u_2) = \mathbf{v} + \mathbf{u}.$$

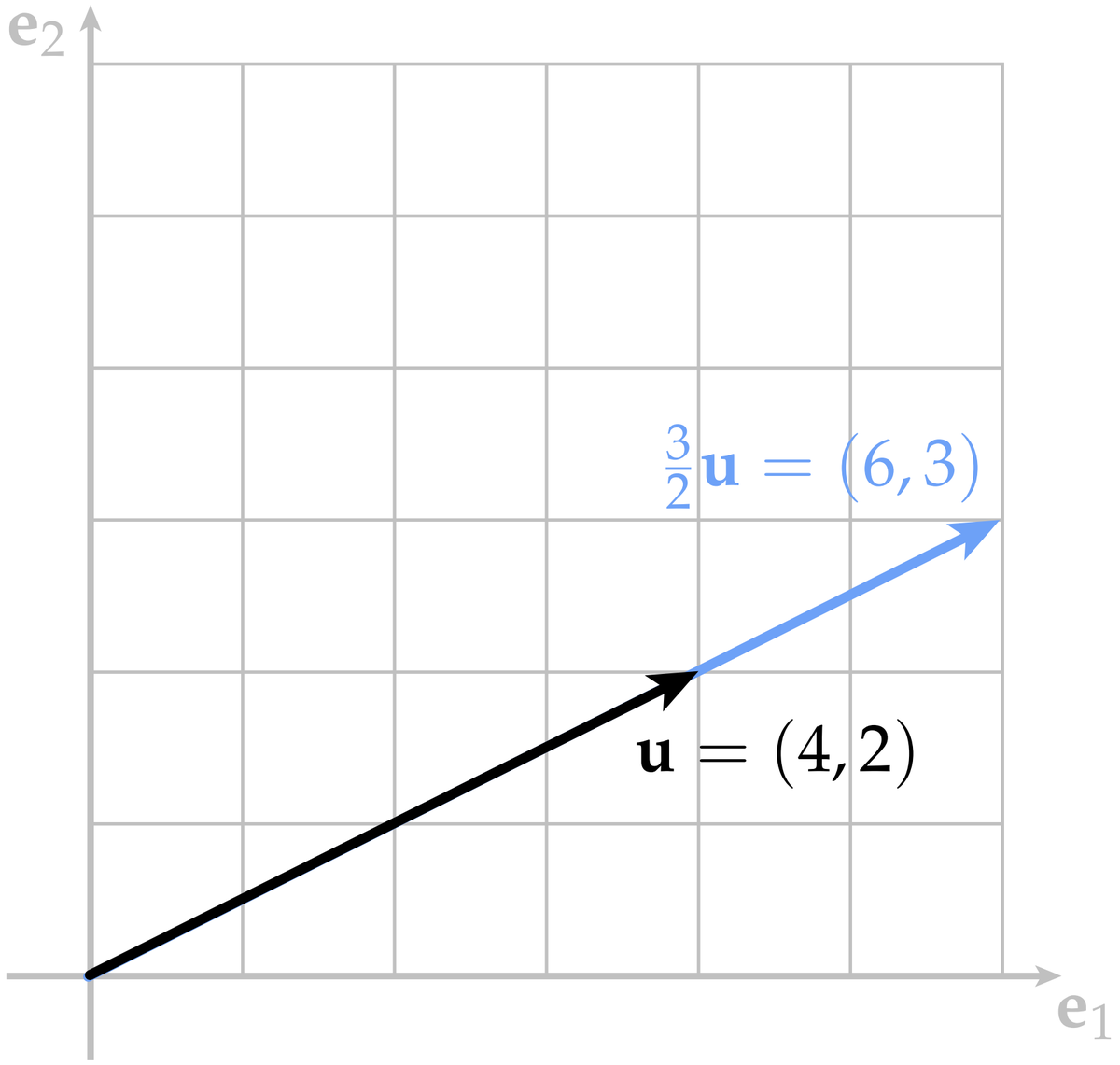
**Turning geometric observations into algebraic rules is convenient for symbolic manipulation & numerical computation.**

**But you should *never* blindly accept a rule given by authority.**

**Always ask: where does this rule come from?**
**What does it mean geometrically?**
**(Can you draw a picture?)**

# Scaling Vectors in Coordinates

- We'd also like to be able to scale vectors using coordinates.
- Any ideas?




$$\frac{3}{2}\mathbf{u}$$

$$= \frac{3}{2}(4,2)$$

$$= (4 \cdot 3/2, 2 \cdot 3/2)$$

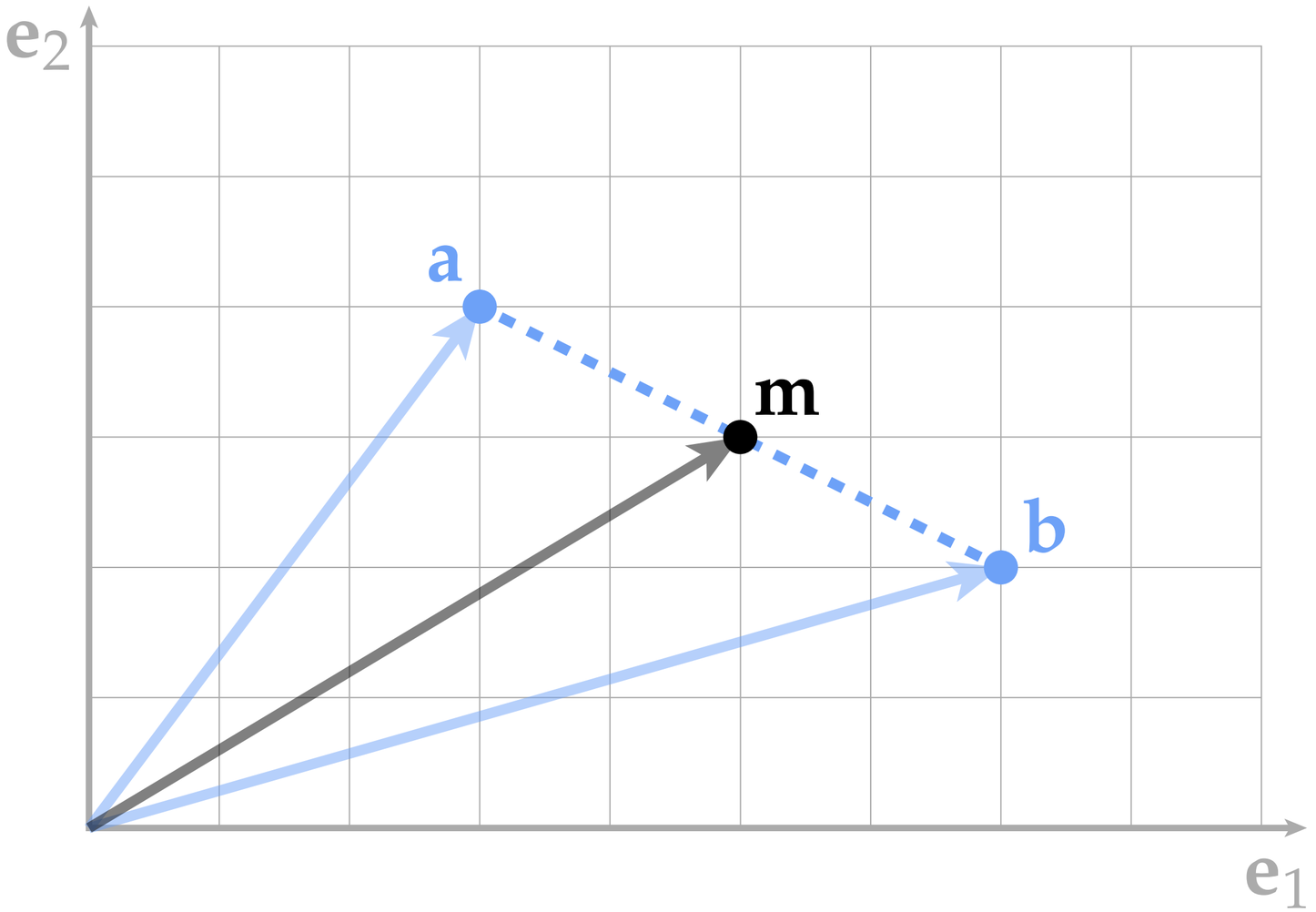
$$= (12/2, 6/2)$$

$$= (6,3)$$

**(From here, check the rest of the properties...)**

# Computing the Midpoint

- As we start to combine vector operations, we build up operations needed for computer graphics.
- E.g., how would I compute the midpoint m of a = (3,4) and b=(7,2)?




$$\mathbf{m} = \tfrac{1}{2}(\mathbf{a} + \mathbf{b})$$

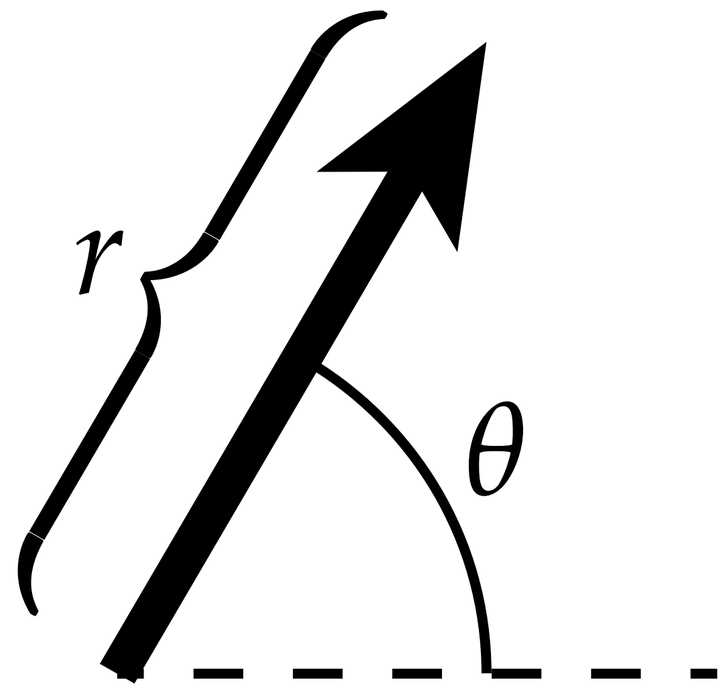
$$= \tfrac{1}{2}((3,4) + (7,2))$$

$$= \tfrac{1}{2}(10,6)$$

$$= (5,3)$$

# Measuring Vectors

- Earlier we asked, “what information does a vector encode?”
- (A: Orientation and magnitude.)
- How do we actually *measure* these quantities?

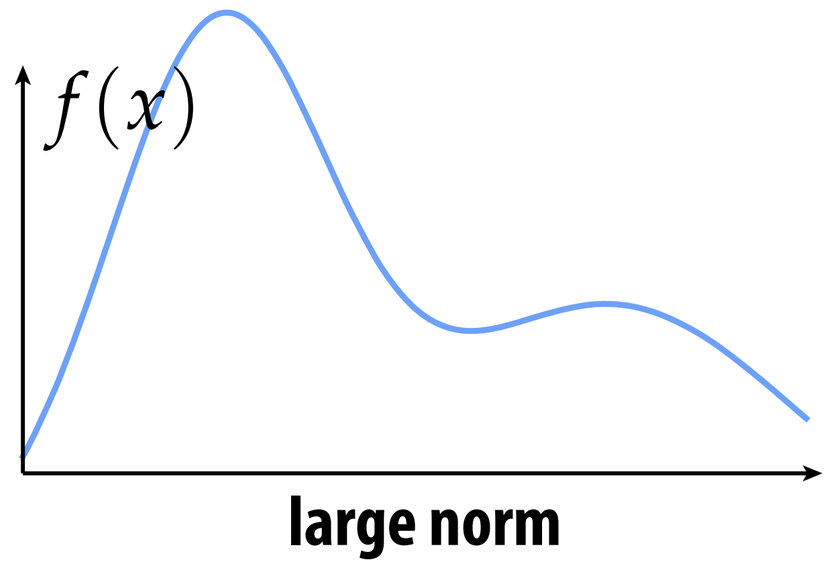


# Norm of a Vector

- Let's start with magnitude—for a given vector v, we want to assign it a number |v| called its length or magnitude or norm.
- Intuitively, the norm should capture how "big" the vector is.

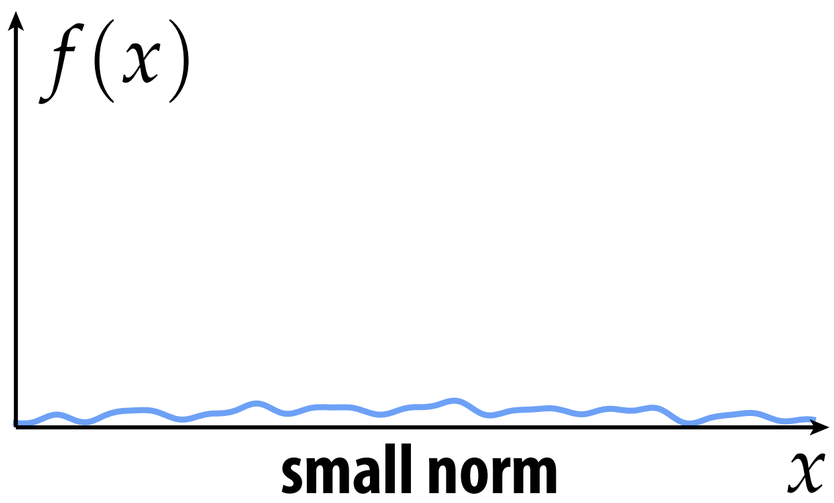
small norm

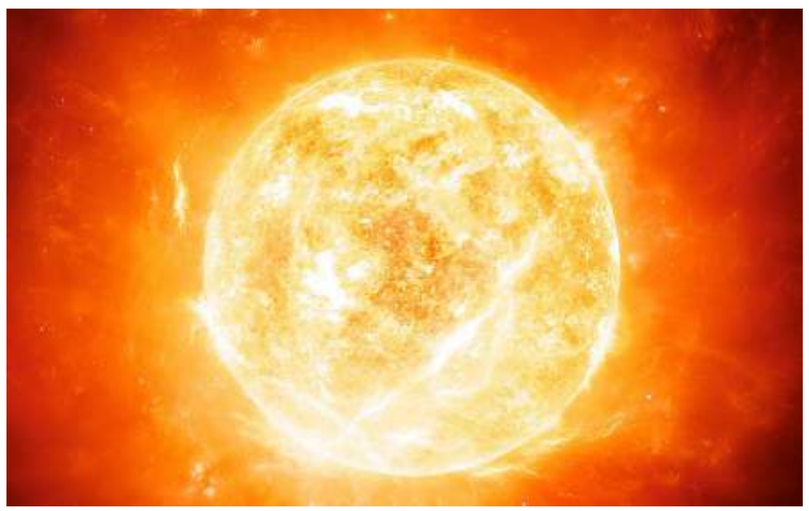
large norm






small norm



large norm

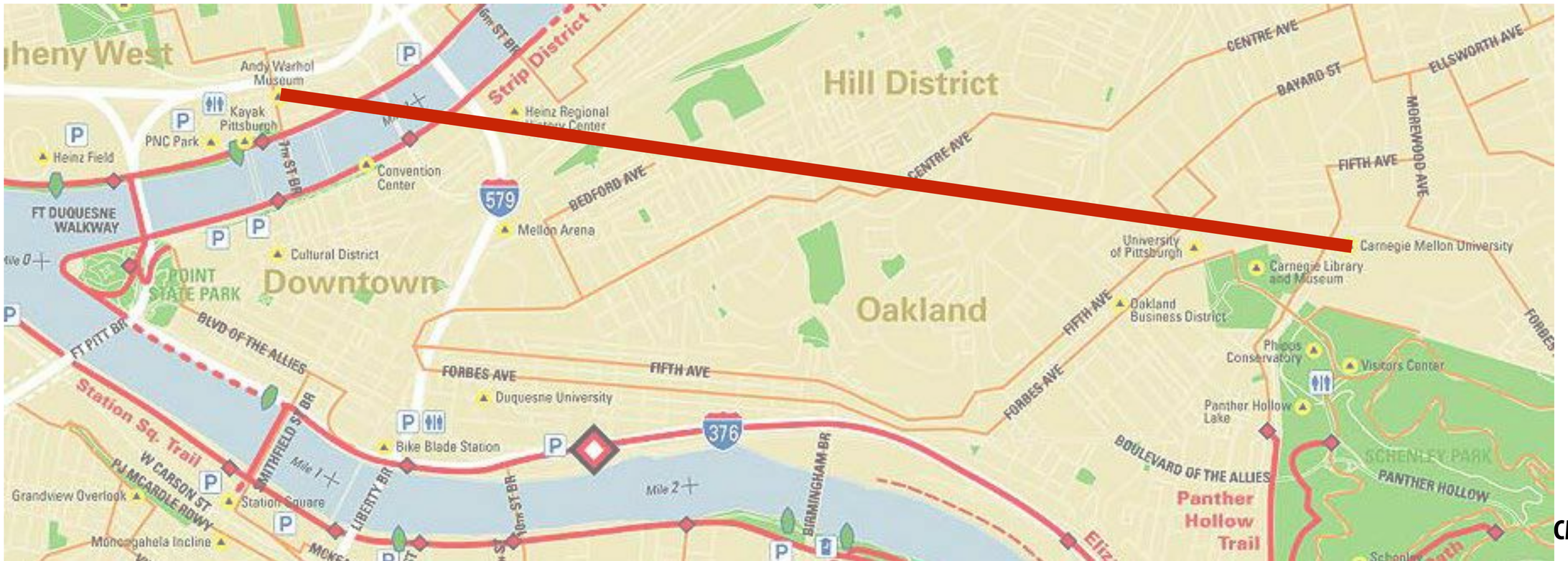
# Natural Properties of Length—Positivity

- What properties might you expect the norm (or length) of a vector to satisfy?
- For one thing, it probably shouldn't be negative!

$$|\mathbf{u}| \geq 0$$

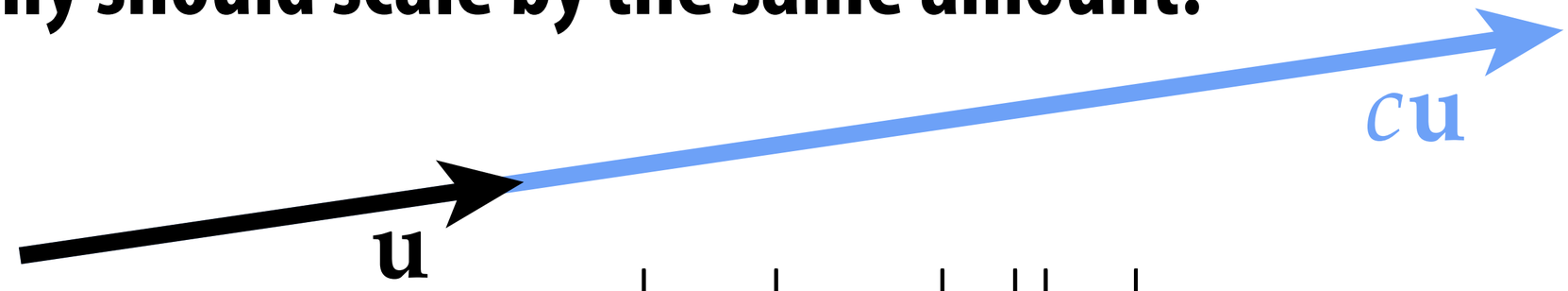
- And probably it should be zero only for the zero vector:

$$|\mathbf{u}| = 0 \iff \mathbf{u} = \mathbf{0}$$



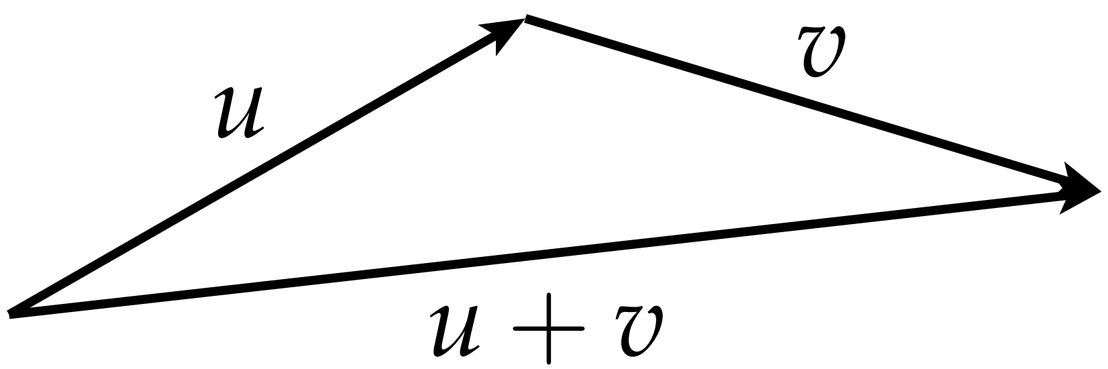
# Natural Properties of Length, Continued

- Also, if we scale a vector by a factor c, its norm (i.e., length) really should scale by the same amount:




$$|c\mathbf{u}| = |c||\mathbf{u}|$$

- Finally, we know that the shortest path between two points is always along a straight line:




$$|\mathbf{u}| + |\mathbf{v}| \geq |\mathbf{u} + \mathbf{v}|$$

- (This final property is sometimes called the "pentagon inequality," since the diagram looks like a pentagon.)

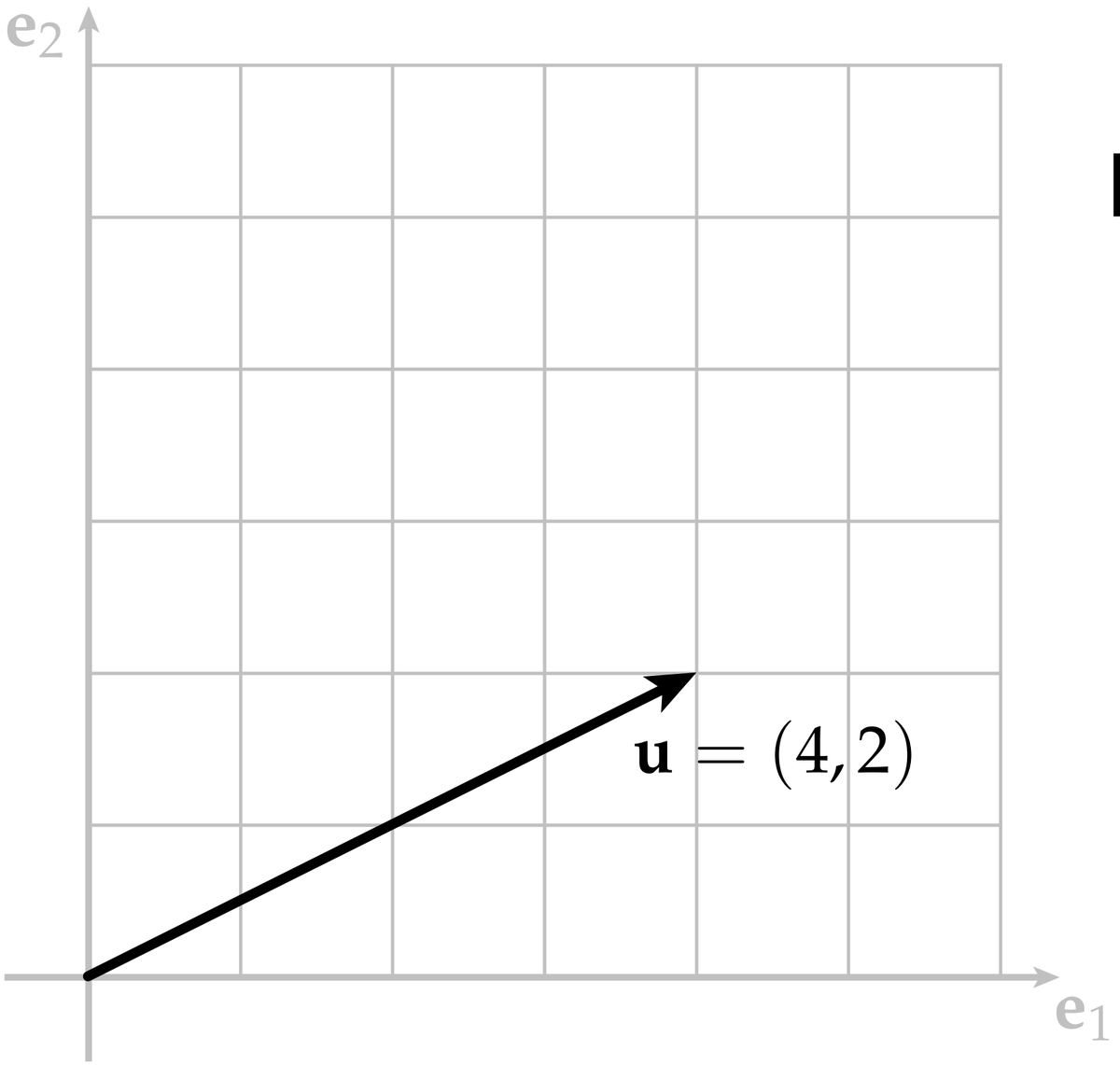
# Norm—Formal Definition

- A norm is any function that assigns a number to each vector and satisfies the following properties for all vectors u, v, and all scalars $a$:
  - $|\mathbf{v}| \geq 0$
  - $|\mathbf{v}| = 0 \quad \iff \quad \mathbf{v} = \mathbf{0}$
  - $|a\mathbf{v}| = |a||\mathbf{v}|$
  - $|\mathbf{u}| + |\mathbf{v}| \geq |\mathbf{u} + \mathbf{v}|$
- But you don't have to take my word for it—for each rule, you now have a concrete geometric picture explaining why this "rule" is there.

# Euclidean Norm in Cartesian Coordinates

- **A standard norm is the so-called *Euclidean norm* of n-vectors:**

$$|\mathbf{u}| = |(u_1, \ldots, u_n)| := \sqrt{\sum_{i=1}^{n} u_i^2}$$




**Example:** $\mathbf{u} = (4, 2)$

$$|\mathbf{u}| = \sqrt{4^2 + 2^2}$$

$$= 2\sqrt{5}$$

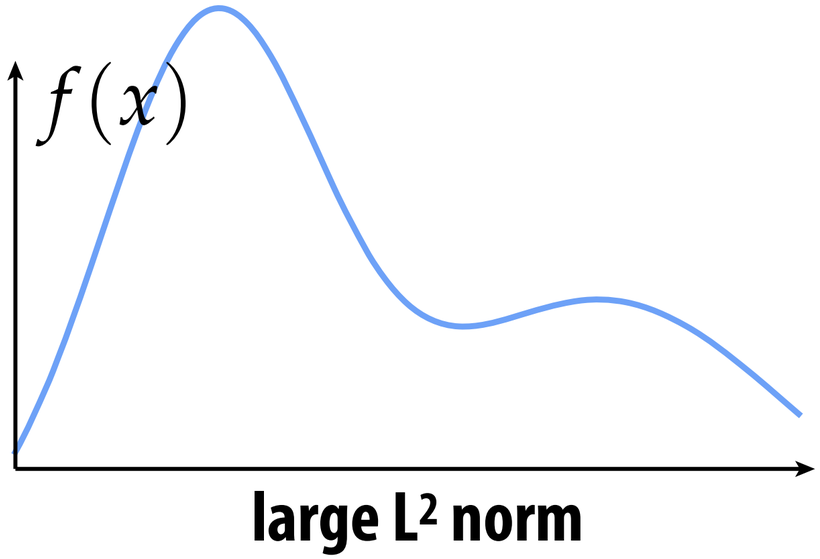
**Q: Does this formula satisfy all the natural, geometric properties of a norm? (Answer in the slide comments!)**

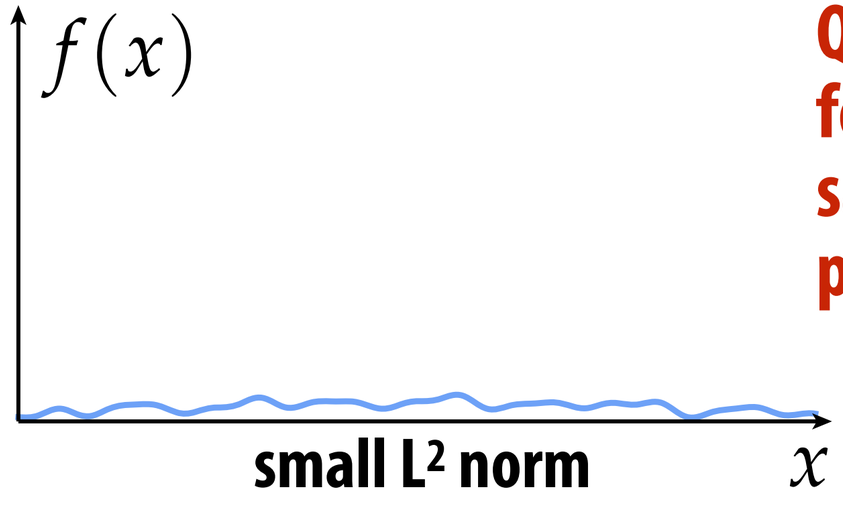
# $L^2$ Norm of Functions

- Less familiar idea, but same basic intuition: the so-called *$L^2$ norm* measures the total magnitude of a function.
- Consider real-valued functions on the unit interval [0,1] whose square has a well-defined integral. The $L^2$ norm is defined as:

$$||f|| := \sqrt{\int_0^1 f(x)^2 \, dx}$$

- Not too different from the Euclidean norm: we just replaced a sum with an integral (which is kind of like a sum…).




Q: Careful—does the formula above *exactly* satisfy all our desired properties for a norm?

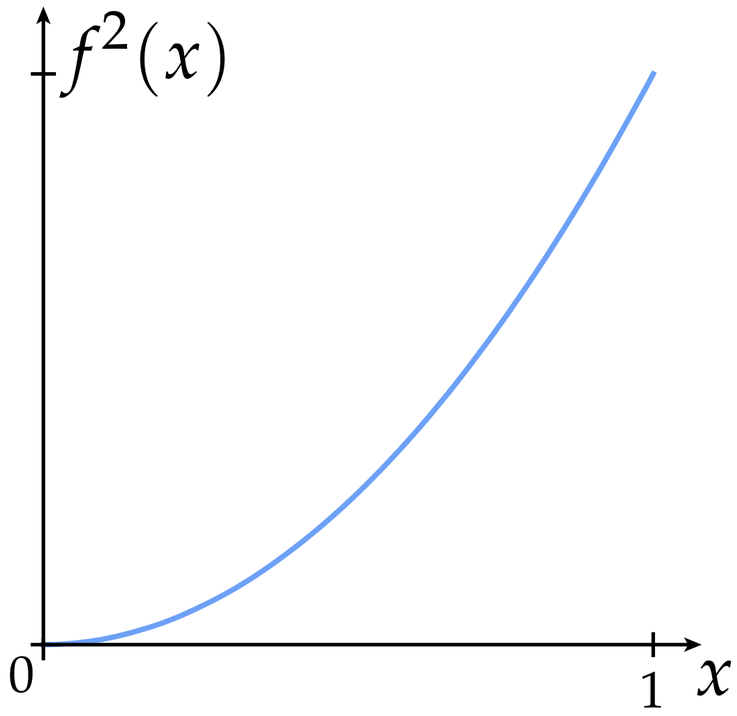
# $L^2$ Norm of Functions—Example

- Consider the function $f(x) := x\sqrt{3}$, defined over the unit interval [0,1].
- Q: What is its $L^2$ norm?
- A:

$$||f|| := \sqrt{\int_0^1 f(x)^2 \, dx}$$

$$||f||^2 = \int_0^1 3x^2 \, dx = \Big[ x^3 \Big]_0^1 = 1^3 - 0^3 = 1$$

$$\Rightarrow ||f|| = \sqrt{1} = 1.$$

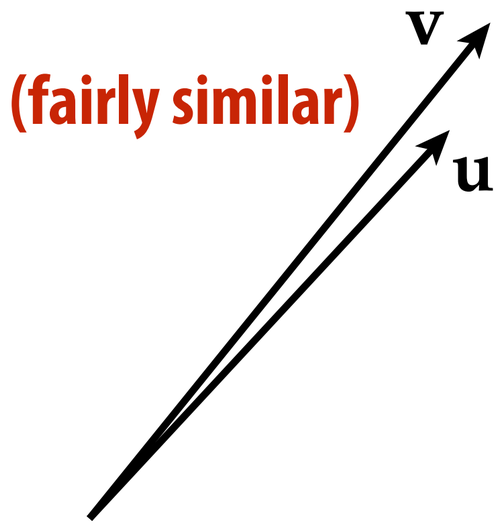


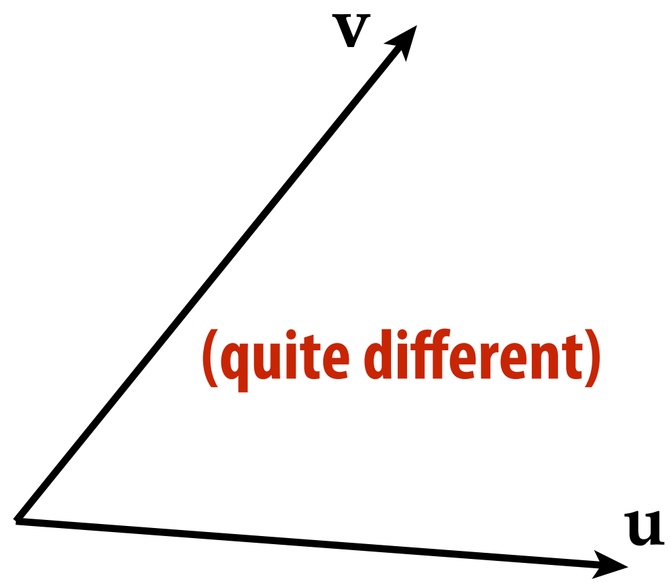

**For clarity we will use $\|\cdot\|$ for the norm of a function, and $|\cdot|$ for the norm of a vector in $R^n$.**

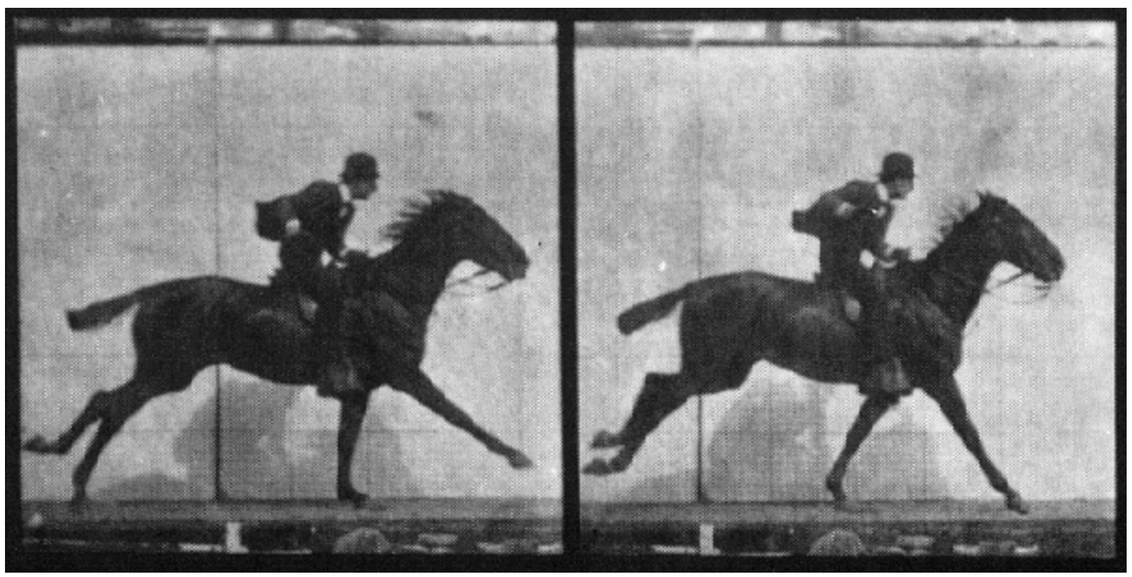
**P.S. Most integrals in graphics are not calculated this way (at least not for more challenging functions, or functions described by data). Later on we'll talk a *lot* about numerical integration.**

# Inner Product—Motivation

- What else can we measure? In addition to magnitude, we said that vectors have *orientation*. Just as norm measured length, inner product measures how well vectors "line up."



(fairly similar)

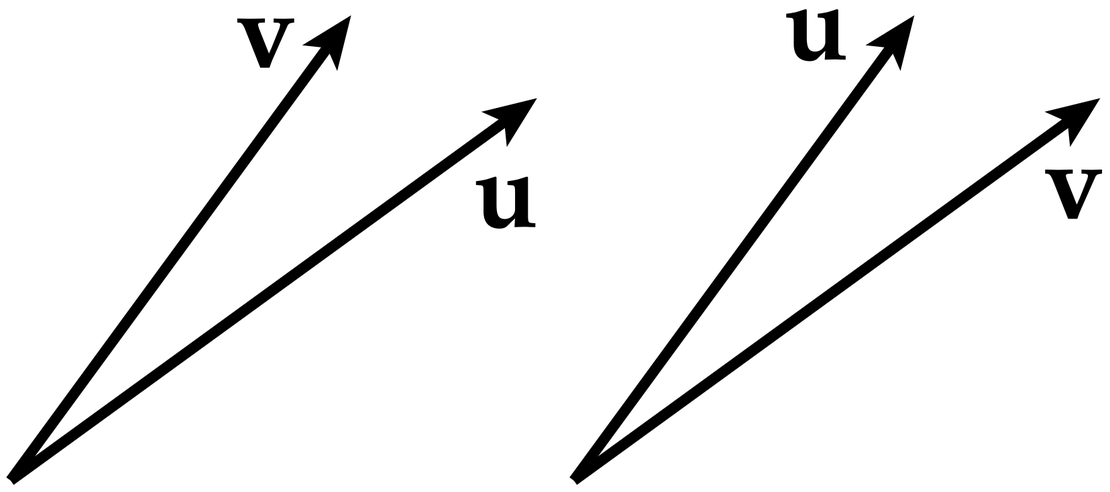


(quite different!)

# Inner Product—Symmetry

- Will write inner product (also sometimes called the scalar product or dot product) using the notation <u,v> (some folks also write it as u•v).
- When measuring the alignment of two vectors u,v, what are some natural properties you might expect?
- One "obvious" property: order shouldn't matter, since u is just as well-aligned with v as v is with u:

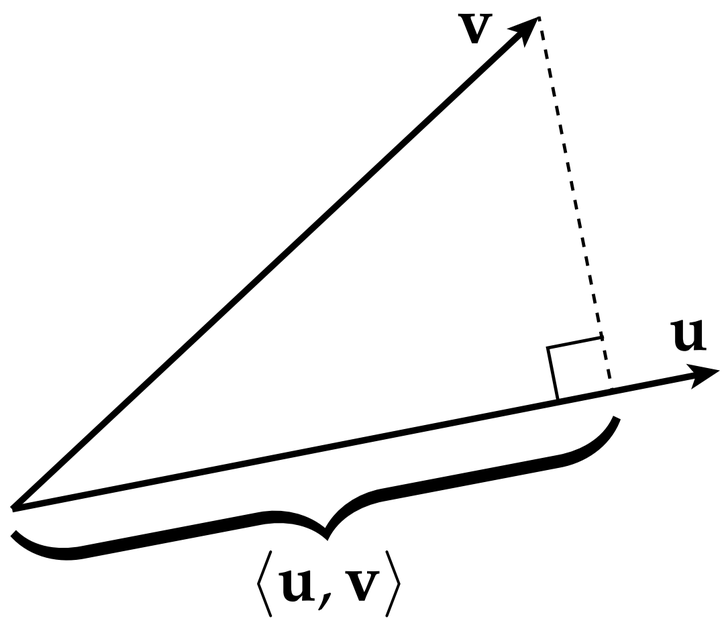
$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$$




- Moreover, simply *re-naming* the vectors should have no effect on how well-aligned they are!

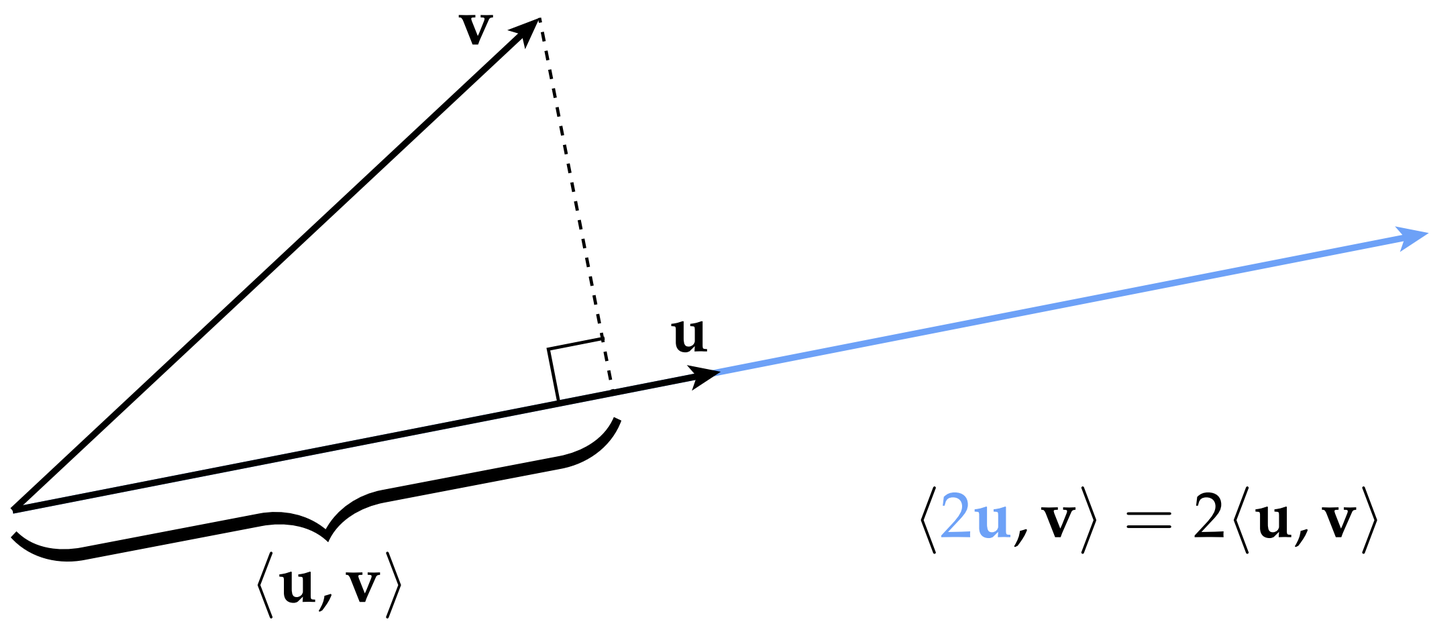
# Inner Product—Projection & Scaling

- For unit vectors $|u|=|v|=1$, an inner product measures the extent of one vector along the direction of the other:




Q: Is this property symmetric?
I.e., is the length of v along u the same as the length of u along v?

- If we scale either of the vectors, the inner product also scales:



# Inner Product—Positivity

- Also, a vector should always be aligned with itself, which we can express by saying that the inner product of a vector with itself should be positive (or at least, non-negative):




$$\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$$

- In fact, if we continue to think of the inner product of a vector as the length of one vector along another then for unit-length vectors we must have

$$\langle \mathbf{u}, \mathbf{u} \rangle = 1$$

- Q: In general, then, what must <u,u> be equal to?
- A: Letting $\hat{\mathbf{u}} := \mathbf{u} / |\mathbf{u}|$, we have

$$\langle \mathbf{u}, \mathbf{u} \rangle = \langle |\mathbf{u}| \hat{\mathbf{u}}, |\mathbf{u}| \hat{\mathbf{u}} \rangle = |\mathbf{u}|^2 \langle \hat{\mathbf{u}}, \hat{\mathbf{u}} \rangle = |\mathbf{u}|^2 \cdot 1 = |\mathbf{u}|^2$$

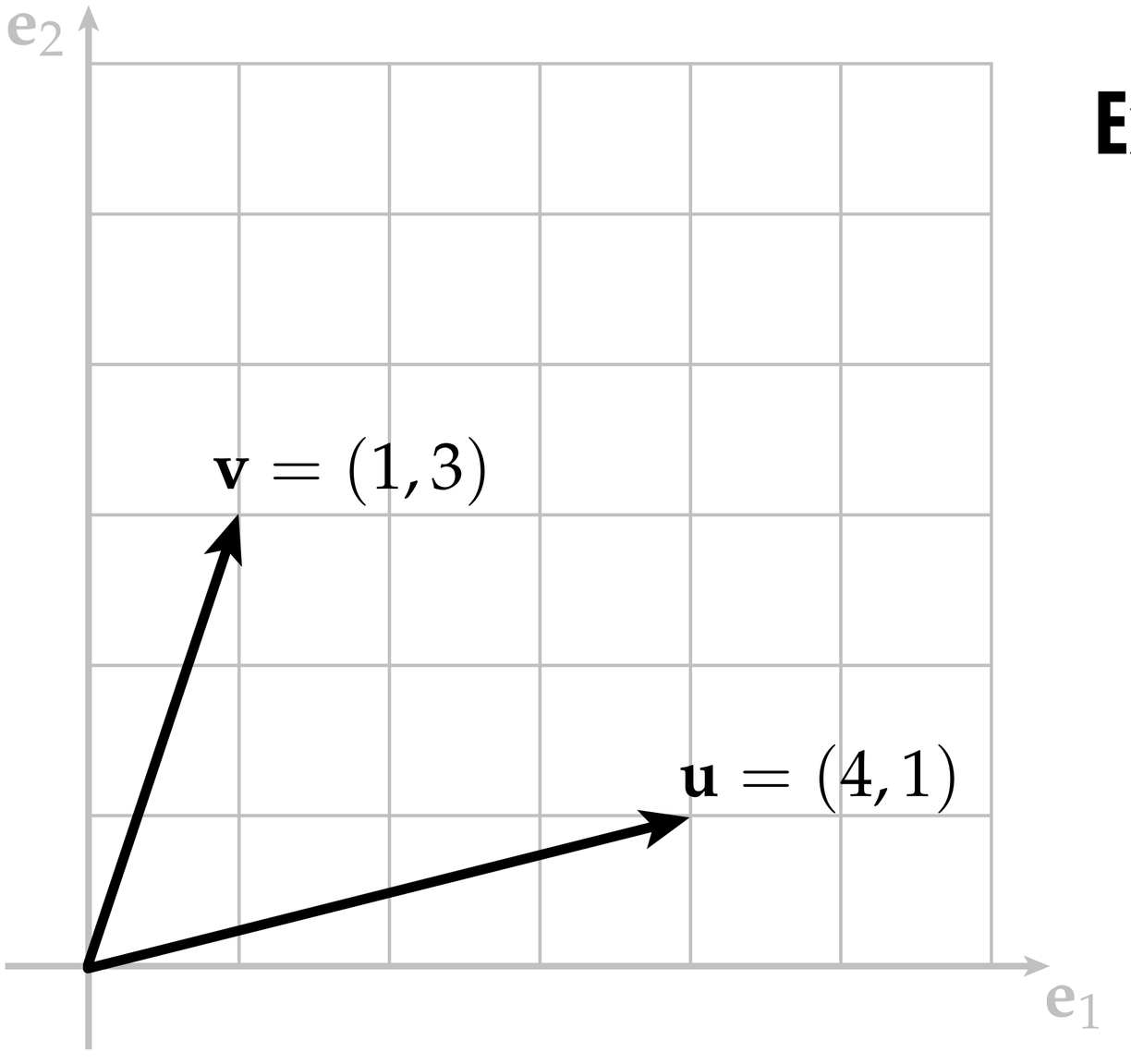
# Inner Product—Formal Definition

- An inner product is any function that assigns to any two vectors u,v a number <u,v> satisfying the following properties:
  - $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
  - $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$
  - $\langle \mathbf{u}, \mathbf{u} \rangle = 0 \quad \Longleftrightarrow \quad \mathbf{u} = \mathbf{0}$
  - $\langle a\mathbf{u}, \mathbf{v} \rangle = a\langle \mathbf{u}, \mathbf{v} \rangle$
  - $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$
- Q: Which of these properties didn't we talk about? Can you argue that they make sense geometrically? (Discuss online!)

# Inner Product in Cartesian Coordinates

- **A standard inner product is the so-called *Euclidean* inner product, which operates on a pair of n-vectors:**

$$\langle \mathbf{u}, \mathbf{v} \rangle = \langle (u_1, \ldots, u_n), (v_1, \ldots, v_n) \rangle := \sum_{i=1}^{n} u_i v_i$$




**Example:**

$$\mathbf{u} = (4,1)$$
$$\mathbf{v} = (1,3)$$

$$\langle \mathbf{u}, \mathbf{v} \rangle = 4 \cdot 1 + 1 \cdot 3 = 7$$

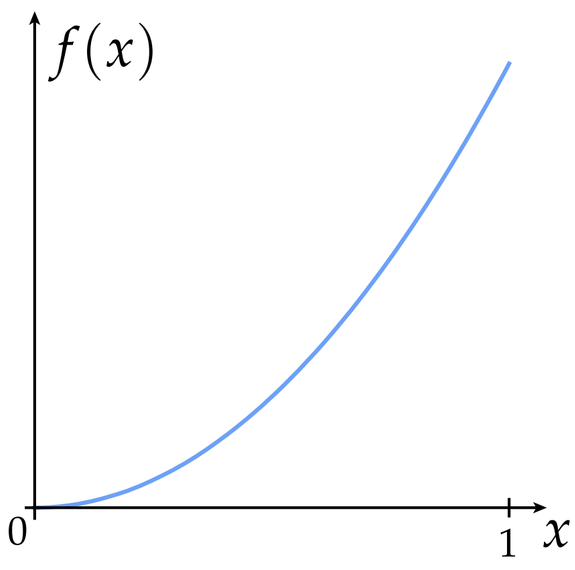
# $L^2$ Inner Product of Functions—Example

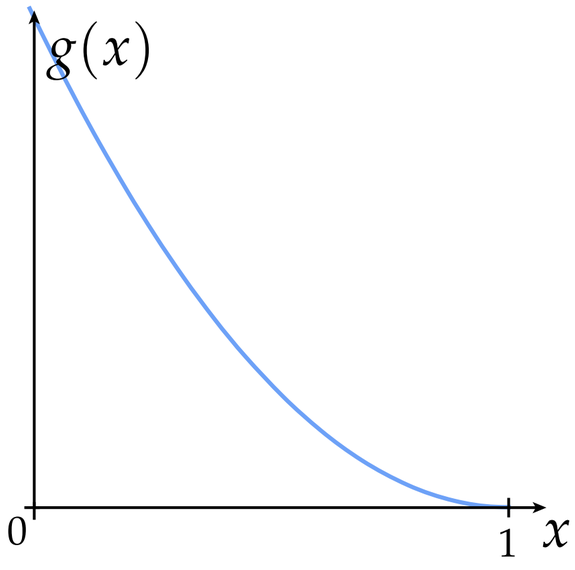
- **Just like we had a norm for functions, we can also define an inner product that measures how well two functions "line up".**
- **E.g., for square-integrable functions on the unit interval:**

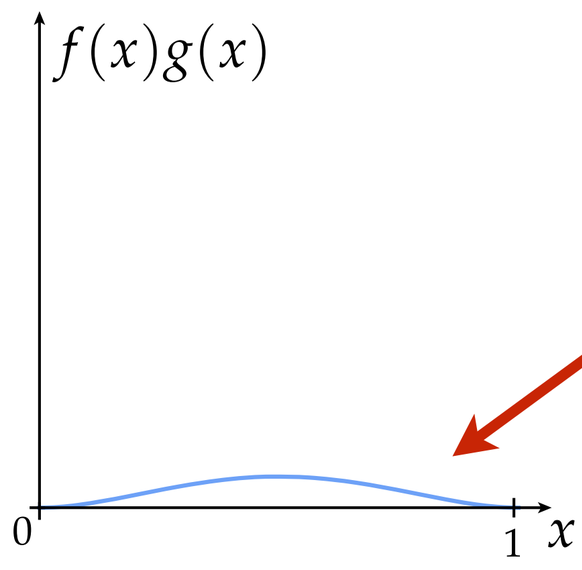
$$\langle\langle f, g \rangle\rangle := \int_0^1 f(x) g(x) \, dx$$

**Example:** $f(x) := x^2, \quad g(x) := (1-x)^2$

$$\langle\langle f, g \rangle\rangle = \int_0^1 x^2 (1-x)^2 \, dx = \cdots = \frac{1}{30}$$



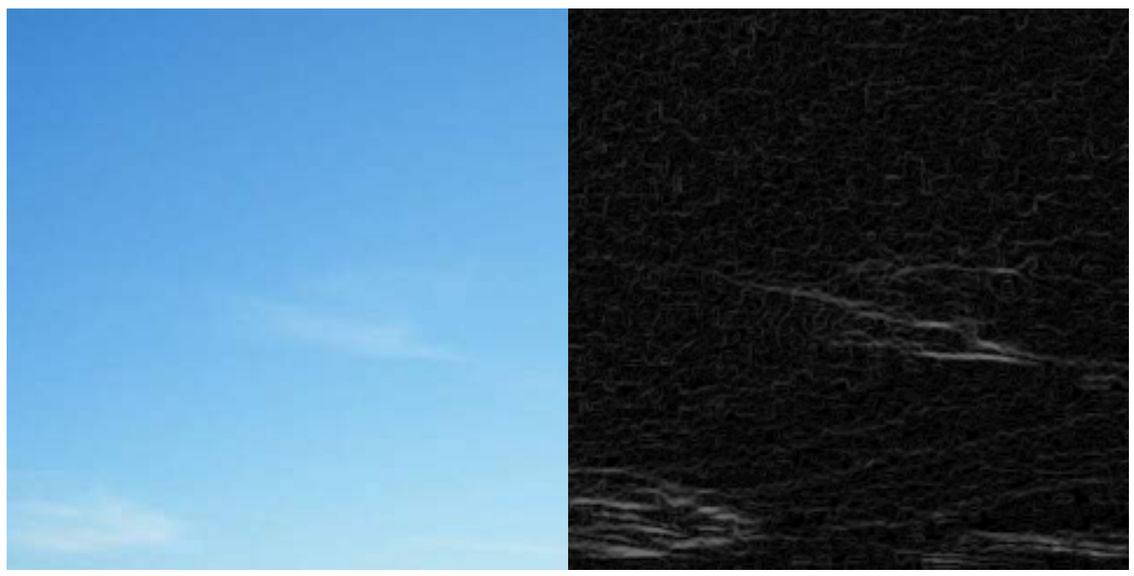

**small number; functions don't "line up" much!**

# Measuring Images, Other Signals?

- Many ways to measure "how big" a signal is (norm) or "how well-aligned" two signals are (inner product).
- Choice of inner product depends on application.
- For instance, suppose we want images with "interesting stuff"
- Might try measuring norm of *derivative* (captures edges):



(dimmer) *LARGER*



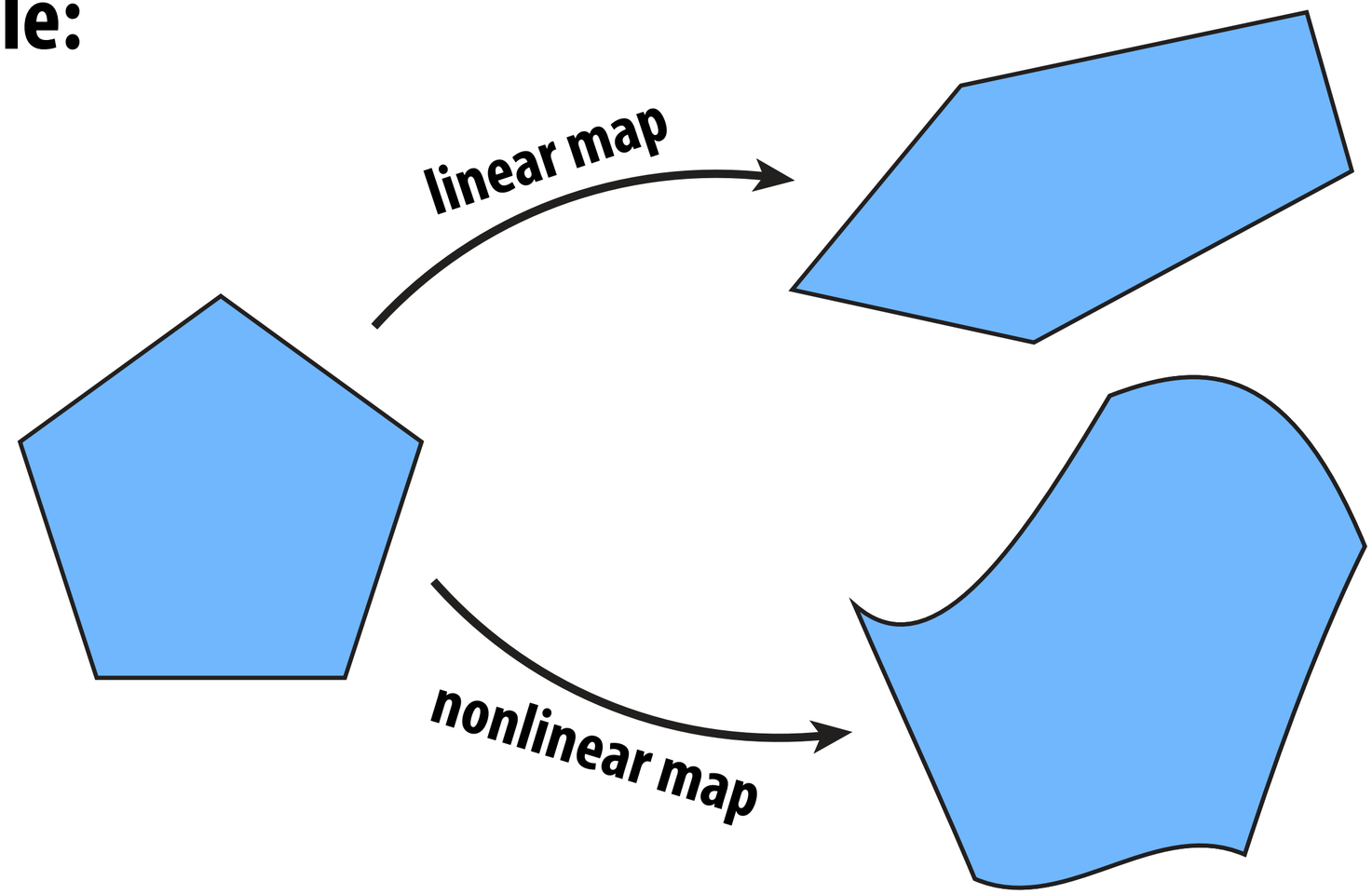
(brighter) *SMALLER*

# Linear Maps

- At the beginning, said linear algebra was study of vector spaces and linear maps between them.
- Have a pretty good handle on vector (and inner product) spaces.
- But what's a linear map? And why is it useful for graphics?
- We'll get to the 1st question in a moment. As for the 2nd question, a few reasons:
  - Computationally, easy to solve systems of linear equations.
  - Basic transformations (rotation, translation, scaling) can be expressed as linear maps. *(Will see this in a later lecture!)*
  - *All* maps can be approximated as linear maps over a short distance/short time. (Taylor's theorem). This approximation is used all over geometry, animation, rendering, image processing...

# Linear Maps—Geometric Definition

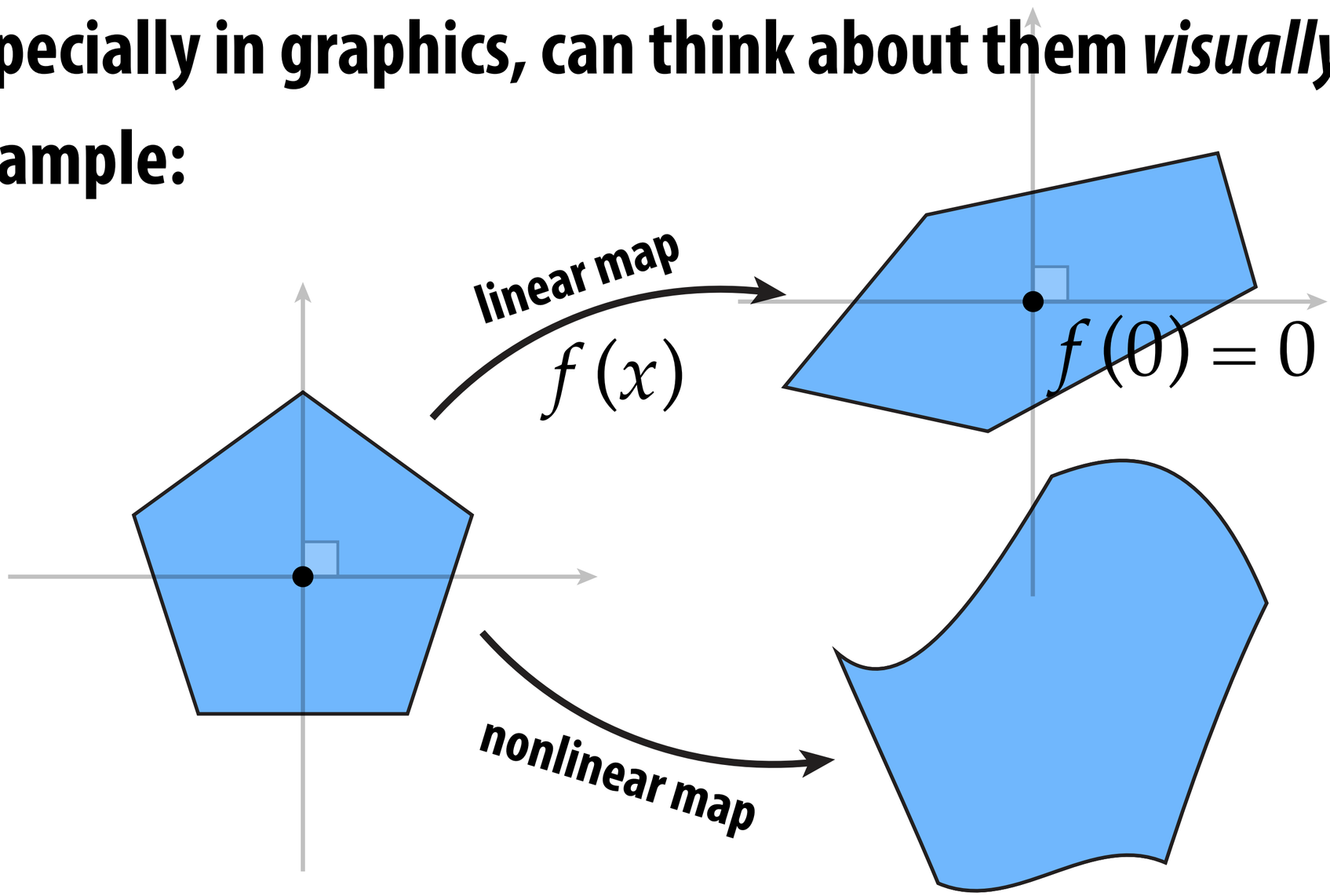
- What is a linear map?
- Especially in graphics, can think about them *visually*.
- Example:




**Key idea: *linear maps take lines to lines**

# Linear Maps—Geometric Definition

- What is a linear map?
- Especially in graphics, can think about them *visually*.
- Example:




**Key idea: *linear maps take lines to lines**

*...while keeping the origin fixed.

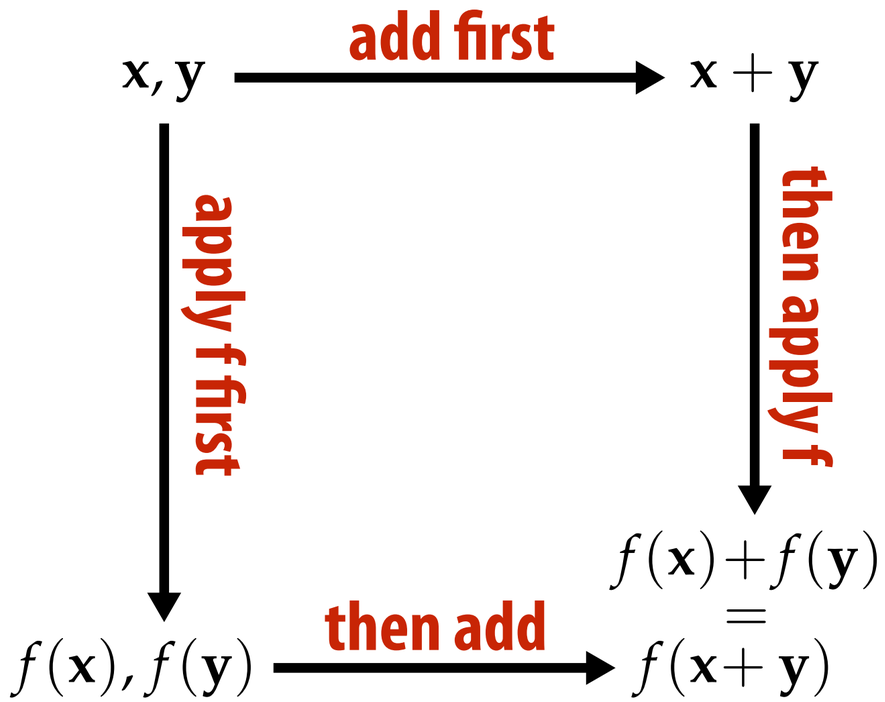
# Linear Maps—Algebraic Definition

- A map f is linear if it maps *vectors to vectors*, and if for all vectors u,v and scalars a we have

$$f(\mathbf{u}+\mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v})$$

$$f(a\mathbf{u}) = af(\mathbf{u})$$

- In other words: if it doesn't matter whether we add the vectors and then apply the map, or apply the map and then add the vectors (and likewise for scaling):

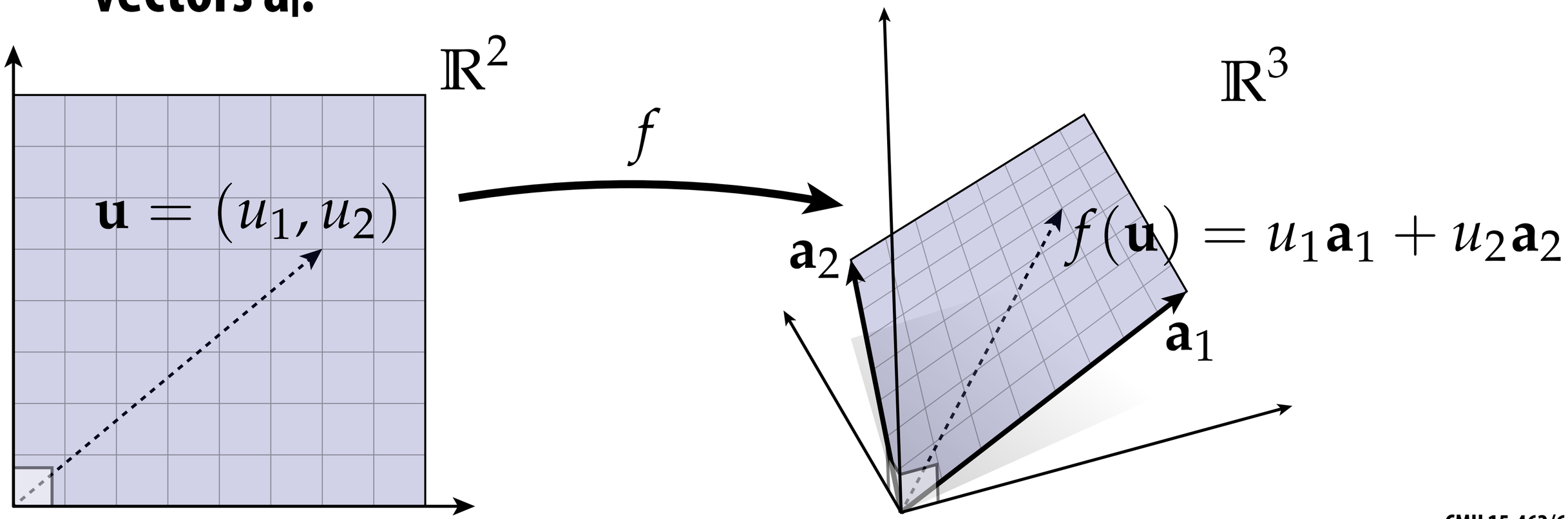


# Linear Maps in Coordinates

- For maps between $\mathbb{R}^m$ and $\mathbb{R}^n$ (e.g., a map from 2D to 3D), we can give an even more explicit definition.
- A map is linear if it can be expressed as

$$f(u_1, \ldots, u_m) = \sum_{i=1}^{m} u_i \mathbf{a}_i$$

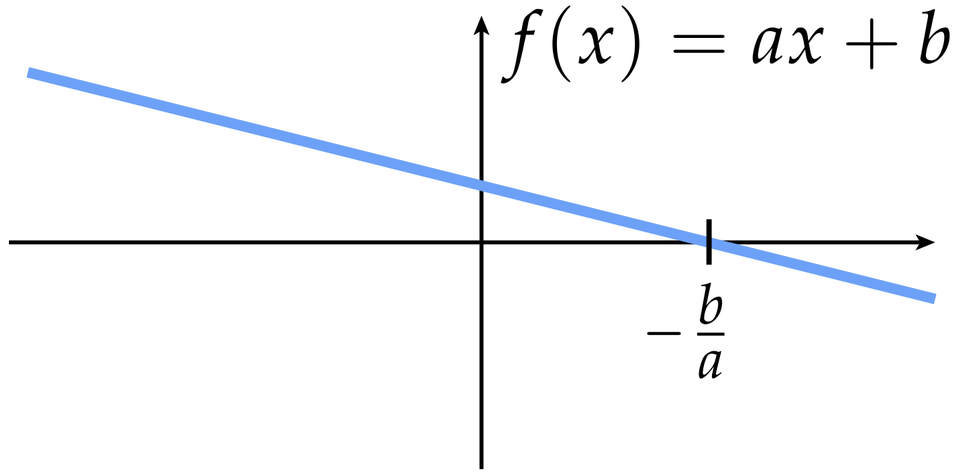
- In other words, if it is a linear combination of a fixed set of vectors $\mathbf{a}_i$:



# Q: Is $f(x) := ax + b$ a linear function?

# Linear vs. Affine Maps

- No! But it's easy to be fooled, since the graph looks like a line:




- However, it's not a line through the *origin*, i.e., f(0) ≠ 0.

- Another way to see it's not linear? Doesn't preserve sums:

$$f(x_1 + x_2) = a(x_1 + x_2) + b = ax_1 + ax_2 + b$$
$$f(x_1) + f(x_2) = (ax_1 + b) + (ax_2 + b) = ax_1 + ax_2 + 2b$$

- This function is called an AFFINE function (not a LINEAR one).

- Later we'll see an important computer graphics magic trick: turn affine functions (e.g., translation) into linear ones via *homogeneous coordinates*.

# More interesting question:

**Q: Is** $f(u) := \int_0^1 u(x)\, dx$ **a linear map?**

**(Think about it—it will be part of your homework!)**

# Span

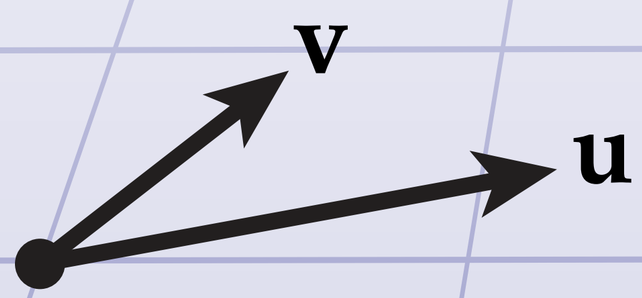
**Q: Geometrically, what is the *span* of two vectors u, v?**

**A: The span is the set of all vectors that can be written as a linear combination of u and v, i.e., vectors of the form**

$$a\mathbf{u} + b\mathbf{v}$$

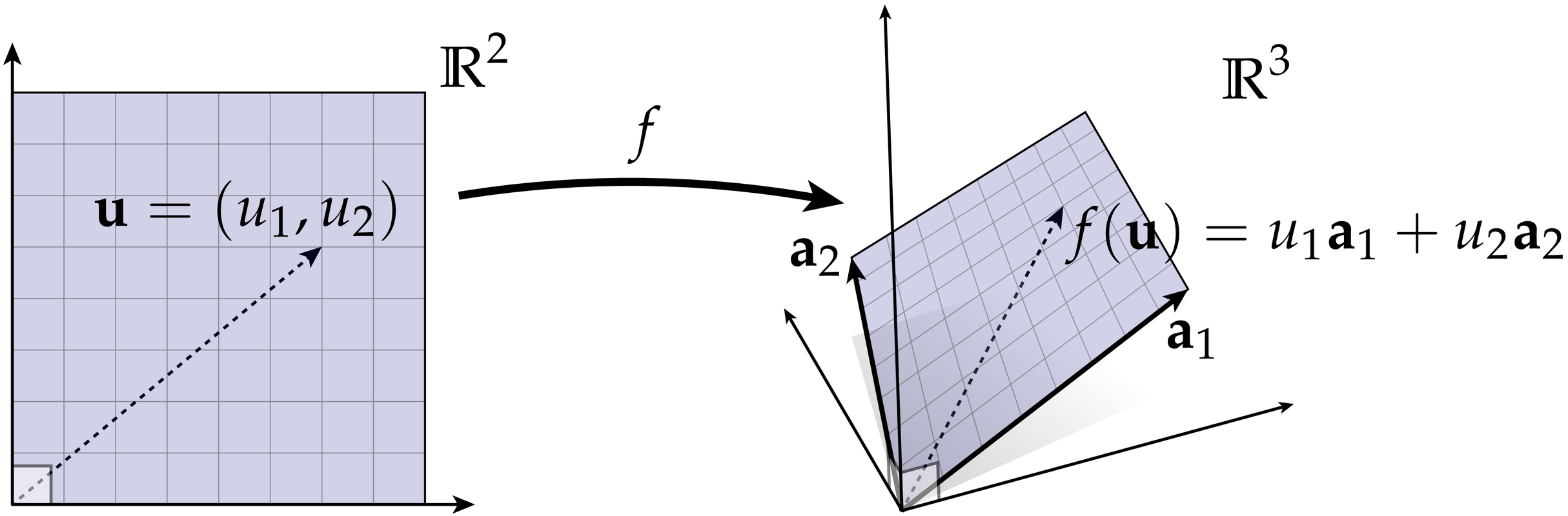
**for any two numbers** a, b**.**

**More generally:** $\operatorname{span}(\mathbf{u}_1, \ldots, \mathbf{u}_k) = \left\{ \mathbf{x} \in V \;\middle|\; \mathbf{x} = \sum_{i=1}^{k} a_i \mathbf{u}_i,\ a_1, \ldots, a_k \in \mathbb{R} \right\}$



# Span & Linear Maps

- **Just a bit of language—can connect "span" and "linear map":**
- ***"The image of any linear map is the span of some collection of vectors."***



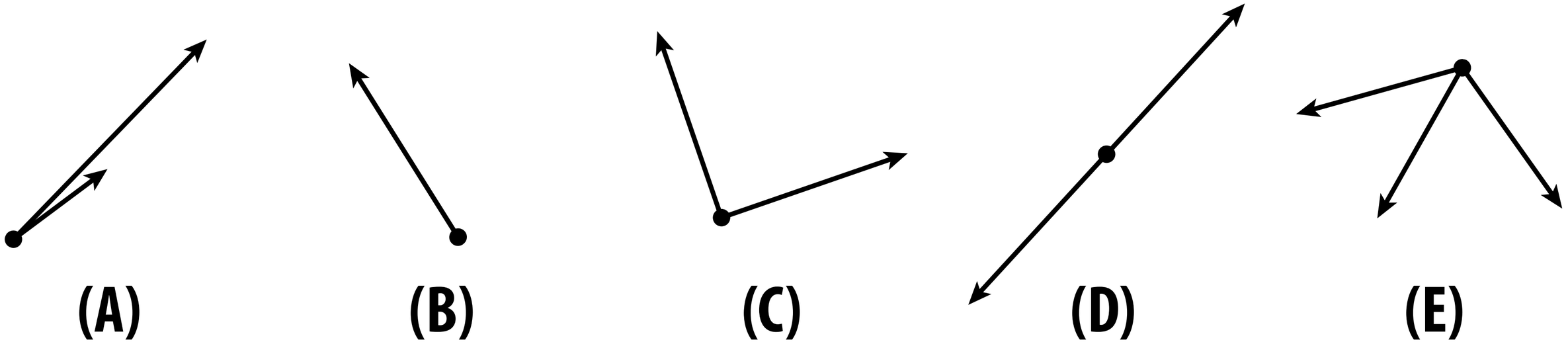

**Q: What's the image of a function?**

# Basis

- Span is also closely related to the idea of a *basis*.
- In particular, if we have exactly n vectors $e_1$, ..., $e_n$ such that

$$\mathrm{span}(\mathbf{e}_1, \ldots, \mathbf{e}_n) = \mathbb{R}^n$$

- Then we say that these vectors are a basis for $R^n$.
- Note: many different choices of basis!
- Q: Which of the following are bases for the 2D plane (n=2)?



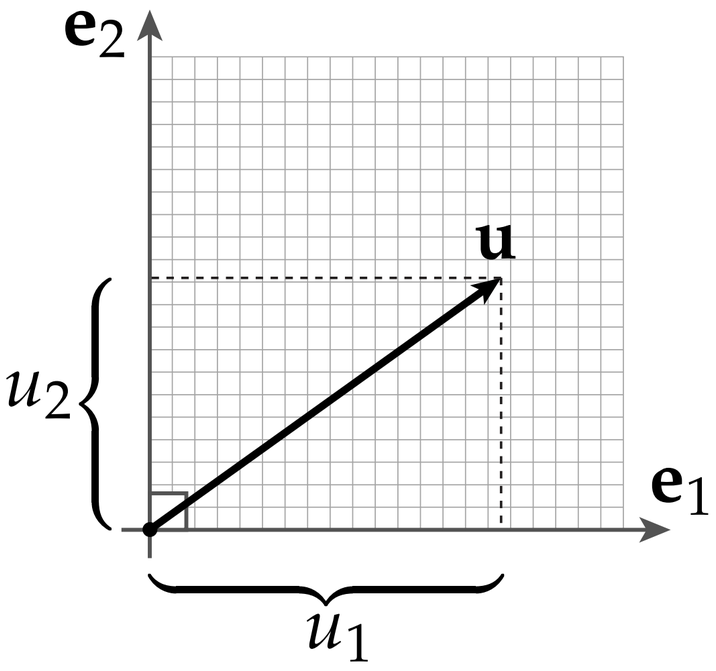
(A) (B) (C) (D) (E)

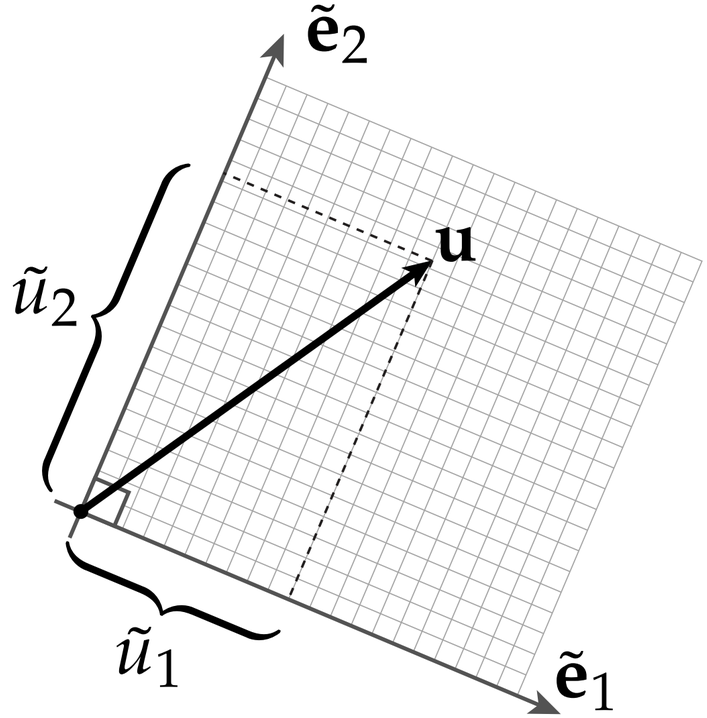
# Orthonormal Basis

- Most often, it is convenient to have to basis vectors that are (i) unit length and (ii) mutually orthogonal.
- In other words, if $\mathbf{e}_1, \ldots, \mathbf{e}_n$ are our basis vectors then

$$\langle \mathbf{e}_i, \mathbf{e}_j \rangle = \begin{cases} 1, & i = j \\ 0, & \text{otherwise.} \end{cases}$$

- This way, the geometric meaning of the sum $u_1^2 + \ldots + u_n^2$ is maintained: it is the length of the vector u.




Common bug: projecting a vector onto a basis that is NOT orthonormal while continuing to use standard norm / inner product.